Modern Algebra I

KNU Math 413

Classnotes

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These notes are for a course taught from Fraleigh's "A First Course in Abstract Algebra" 7th edition. We refer to this book as 'the text'. These notes do not add much to the text, except sometimes a different perspective and a culling of material. We are careful to frequently reference the text for proofs and problems so that the notes can only be used with the text. You will find that many of my problems are just examples from the book.

0 Sets and Relations

A proper mathematical definition of a 'set' takes some work, but usually it is enough to distinguish it from the word 'collection' which in an non mathematical sense is the same thing. A *set* S is an unordered collection of distinct *elements*. The elements can be anything, but are generally similar mathematical objects. If an object a is an *element*, or *member*, of S, then we write $a \in S$. We use 'braces' or 'curly brackets' $\{\quad\}$ to designate that a list of elements is a set, and have several ways to imply elements in the brackets. The set S of integers from 1 to 7 can be written as

$$S = \{1, 2, 3, 4, 5\} \quad \text{or} \quad S = \{1, \ldots, 5\} \quad \text{or} \quad S = \{a \mid a \in \mathbb{Z}, 1 \leq a \leq 5\}$$

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A set A is a *subset* of B, written, $A \subseteq B$ or $B \supseteq A$

$$(x \in A) \Rightarrow (x \in B).$$

This cool thick arrow '$\Rightarrow$' is read 'implies'. This symbolic sentence means 'if x is in A, then x is in B'. A is a *proper subset* of B, written $A \subset B$ if $A \subseteq B$ but $A \neq B$.

You should be familiar with the following sets that are explained on page 3 of the text:

$$\emptyset \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \qquad \mathbb{Z}^* \subset \mathbb{Q}^* \subset \mathbb{R}^* \subset \mathbb{C}^* \qquad \mathbb{Z}^+ \subset \mathbb{Q}^+ \subset \mathbb{R}^+.$$

The *union* of two sets A and B is $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$ and the *intersection* is $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.

A *function* $f : A \to B$ from a set A to a set B is a rule that assigns an element $f(a)$ of B to every element a of A. For a subsets $A' \subseteq A$ and $B' \subset B$ we write

$$f[A'] = \{f(a) \mid a \in A'\} \qquad \text{and} \qquad f^{-1}[B'] = \{b \mid f(b) \in B'\}.$$

A is the *domain* of f and B is the *codomain*. The set $f[A] \subseteq B$ is its *range*. A function $f : A \to B$ is *injective* or *one-to-one* if $f(x) = f(y) \Rightarrow x = y$,

and is *surjective* or *onto* if $f[A] = B$. If it *bijective* if it is both injective and surjective. A function $f : A \to B$ has an inverse function $f^{-1} : B \to A$ such that $f^{-1}(b) = a$ if $f(a) = b$ if and only if f is bijective.

The *cardinality* $|A|$ of a finite set A is the number of elements it has. Defining $|A|$ is more difficult if A is infinite, but we can say that two sets A and B have the same cardinality if there is a bijection (bijective function) from A to B. (Actually, this is how we define cardinality in an axiomatic set theory class.).

Example 0.1. Elements of a set are unique so $|\{1, 1, 1, 2, 3\}| = |\{1, 2, 3\}| = 3$.

Problem 0.1. Read the text for the proofs that $|\mathbb{Z}^+| = |\mathbb{Z}| = |\mathbb{Q}|$ and the proof that $|\mathbb{Z}| \neq |\mathbb{R}|$.

Problem 0.2. If $f : \mathbb{R} \to \mathbb{R}$ is defined $f(x) = 2x$, what are $f[\mathbb{Z}]$ and $f^{-1}[\mathbb{Z}]$?

Relations

The *(cartesian) product* of sets A and B is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

A *relation* on a set S is a subset $\mathscr{R} \subset S \times S$. Two elements a and b of $\mathscr{R}$ are *related* if $(a, b) \in \mathscr{R}$. We also write this as $a\mathscr{R}b$.

You are probably familiar with the 'equality' relation $=$ and the 'less than' relation $\leq$ on $\mathbb{Z}$. The less than relation is hard to write right now without using the symbol $\leq$, but the equality relation on $\mathbb{Z}$ can be defined as

$$\{(a, a) \mid a \in \mathbb{Z}\}.$$

A relation $\mathscr{R}$ on A is

- *reflexive* if $(a, a) \in \mathscr{R}$ for all $a \in \mathscr{R}$,
- *symmetric* if $(a, b) \in \mathscr{R}$ implies $(b, a) \in \mathscr{R}$, and
- *transitive* if $(a, b) \in \mathscr{R}$ and $(b, c) \in \mathscr{R}$ imply $(a, c) \in \mathscr{R}$.

Problem 0.3. For each subset of { reflexive, symmetric, transitive } come up with a relation that has those properties, but none of the other properties in the set. Try to use relations that are as natural as possible.

A relation $\mathscr{R}$ that is reflexive and transitive is a *pre-order*, if it is anti-symmetric– for all $a \neq b$ in A, $(a, b) \in \mathscr{R}$ implies $(b, a) \notin \mathscr{R}$– then it is a *partial order*. Preorders and partial orders are often represented by $\leq$ or something similar. A relation that is reflexive, symmetric and transitive is an *equivalence relation*, such relations are usually represented $\sim$ or $=$ or $\equiv$ or $\cong$.

Problem 0.4. Recall that a divides b if there is some $z \in \mathbb{Z}$ such that $ca = b$. Define the divides relation $|$ on $\mathbb{Z}$ by $a|b$ if a divides b. Show that the divides relation is a partial order.

Problem 0.5. Show that the modulo n relation $\equiv_n$ on $\mathbb{Z}$ defined by $a \equiv_n b$ if $n|b-a$ is an equivalence relation.

Given an equivalence relation $\sim$ on a set A, the *equivalence class* of a is

$$\overline{a} := \{b \in A \mid a \sim b\},$$

and the set

$$A/_\sim := \{\overline{a} \mid a \in A\},$$

called 'A modulo $\sim$', is the set of these equivalence classes.

Example 0.2. We have $\mathbb{Z}_{\equiv_n} = \{\overline{0}, \overline{1}, \ldots, \overline{n-1}\}$. We usually write this as $\mathbb{Z}_n$.

The set $A/_\sim$ has the following property, which is exactly the property that a set $\mathscr{P}$ of non-empty subsets of a set A must have to be a *partition* of A: every element of A is in exactly one of the sets in $\mathscr{P}$.

Problem 0.6. Show that the set $\mathscr{P} = \{\overline{a} \mid a \in A\}$ is indeed a partition, and that a partition $\mathscr{P}$ of A defines an equivalence relation $\sim$ on A by setting $a \sim b$ if there is some set P in $\mathscr{P}$ with $a, b \in P$.

Problems from the Text

Section 0: 1-11, 15, Read definition before 16, 16-18, Read instructions before 29, do 31,32

1 Introduction and Examples

The main object of this class is a group. A group is a set G with an operation

$$*\colon G \times G \to G$$

satisfying certain properties. Addition and multiplication are examples of group operations. Group theory is the study of what properties various operations have on various sets. The usual real multiplication

$$\times\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

has the property that for every c and b there is an a such that $a \times b = c$. This tells us that $\mathbb{R}$ has inverses under multiplication. The set $\mathbb{Z}$ under the same operation does not have inverses. This leads to some vast differences in the algebraic structure of $\mathbb{R}$ and $\mathbb{Z}$.

Note

We almost always write $\cdot$ for $\times$, writing $a \cdot b$ rather than $a \times b$, but we use the '$\times$' in

$$\times\colon G \times G \to G$$

because $\cdot\colon$ looks like a triangle.

We start by recalling the complex numbers; their multiplication naturally forms interesting groups that we often develop with addition on the integers.

Recall that i is the number such that $i^2 = 1$. It is not in our usual numberline. The complex numbers are the set of numbers

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}.\}$$

Viewing $r \in \mathbb{R}$ as the complex number $r + 0i$, we see that $\mathbb{C}$ contains $\mathbb{R}$. Representing the real number $z = a + bi$ by the point (a, b) in the xy-plane we get a representation of $\mathbb{C}$, the x-axis corresponds to the real numbers $\mathbb{R}$. Usually we call the y-axis, the *i-axis* or the *imaginary axis*, and call this the *complex plane*.

Addition and multiplication on $\mathbb{C}$ are defined

$$(a + ib) + (a' + ib') = (a + a') + i(b + b')$$
$$(a + ib) \cdot (a' + ib') = aa' + i(ba' + ab') + i^2 bb' = (aa' - bb') + i(ba' + ab')$$

Problem 1.1. Compute $(3 + 4i) + (-1 + i)$ and $(2 + i)(2 - i)$ and $7(3 + i)$.

We can check for numbers z_1, z_2 and z_3 of $\mathbb{C}$ that these operations satisfy such identities as

- $z_1 + z_2 = z_2 + z_1$ and $z_1 z_2 = z_2 z_1$,
- $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$ and $z_1(z_2 z_3) = (z_1 z_2) z_3$, and
- $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$.

The absolute value $|z|$ of a complex number z is its distance in the complex plane from the point $0 = 0 + i0$. For $z = a + ib$ we compute it, by Pythagoras, as

$$|z| = |a + ib| = \sqrt{a^2 + b^2}.$$

The point z in the complex plane can also be described in terms of the angle θ that the line through it and 0 makes with the real axis.

$$z = |z|(\cos\theta + i\sin\theta) = |z|e^{\theta},$$

where the representation on the right uses Euler's well known equation $e^{i\theta} = \cos\theta + i\sin\theta$.

This polar representation of complex numbers really elucidates their multiplicative structure. As

$$|z_1|e^{i\theta_1} \cdot |z_2|e^{i\theta_2} = |z_1||z_2|e^{i(\theta_1 +_{2\pi} \theta_2)}.$$

What is this '$+_{2\pi}$'? It is *addition modulo* 2π. For real numbers r and r' we define $r +_{2\pi} r'$ as the unique value s in $[0, 2\pi)$ such that $r + r' + c(2\pi) = s$ for some integer c. In the same way, we can define addition $+_r$ modulo r on $\mathbb{R}$ for any any number r. If r is an integer, the operation is *closed* on $\mathbb{Z}$, meaning that if a and b are in $\mathbb{Z}$, then so is $a +_r b$.

Problem 1.2. Describe this formula of multiplication of complex numbers with a figure in the complex plane.

With this, we can solve equations.

Problem 1.3. The equation $x^2 = -1$ has no solutions in $\mathbb{R}$, and i was defined to be a solution in $\mathbb{C}$. Are there any other solutions in $\mathbb{C}$?

Solution

Sure. $-i$. Any more? No. No fair using the fundamental theorem of algebra though, we prove that later. Let $z = re^{i\theta}$ be a solution. Then $e^{i\pi} = -1 = z^2 = r^2e^{i2\theta}$ so $r^2 = 1$ and $2\theta = \pi$. As $r \in \mathbb{R}^+$, the first equation gives $r = 1$, and as $\theta \in [0, 2\pi)$ the second gives $\pi/2$ and $3\pi/2$. The solutions are $e^{i\pi/2}$ which is i, and $e^{i3\pi/2}$ which is $-i$.

Problem 1.4. Find all solutions, in $\mathbb{C}$, to $z^4 = -16$.

It is not hard to see that the equation $z^n = 1$ has exactly n solutions in $\mathbb{C}$, the nth *roots of unity*

$$U_n := \{e^{ic\frac{2\pi}{n}} \mid n = 0, 1, \ldots, n-1\} = \{\zeta_n^0, \zeta_n^1, \zeta_n^2, \ldots, \zeta_n^{n-1}\},$$

where $\zeta_n = e^{i\frac{2\pi}{n}}$. (This ζ is the greek letter 'zeta'.) Multiplication in U_n looks a lot like addition modulo n:

$$\zeta_8^2\zeta_8^3 = \zeta_8^{2+3} = \zeta^5 \qquad \text{and} \qquad \zeta_8^7\zeta_8^5 = \zeta_8^{7+_8 5} = \zeta_8^4.$$

Indeed we will see that addition modulo n is a group operation on the set $\mathbb{Z}_n$ defined in Example 0.2. It will be isomorphic to the group U_n of nth roots of unity under multiplication.

Problems from the Text

Section 1: 1-21 (Many problems are similar, do enough that you know you can do them).

2 Binary Operations

A *binary operation* $*$ on a set S is a function

$$* : S \times S \to S.$$

Often, if the operation is a symbol such as $*$, we write $a * b$ for $*(a, b)$.

Common examples are the binary operations $+$ and $\cdot$ defined on $\mathbb{Z}$ or $\mathbb{R}$ or $\mathbb{C}$. Matrix addition $+$ is a binary operation on the set of $m \times n$ matrices over $\mathbb{R}$, but it is not a binary operation on the set of real matrices– it is not defined if the matrices have different dimensions.

A binary operation $*$ on a set S is *closed* on a subset $T \subset S$ if for all t, t' in T, $t * t'$ is also in T. We also say that T is *closed under* $*$.

Example 2.1. The binary operation of division on $\mathbb{R}^+$ is not closed under the subset $\mathbb{Z}^+$ of $\mathbb{R}^+$. Why not?

Problem 2.1. Show that the binary operation $\min : \mathbb{R} \times \mathbb{R} \to \mathbb{R} : (a, b) \mapsto \min(a, b)$ is closed on any subset of $\mathbb{R}$. Can you think of another operation on $\mathbb{R}$ with this property?

Problem 2.2. Find a binary operation on $\mathbb{R}$ that is closed on a subset S of $\mathbb{R}$ if and only if S contains the element 1.

A useful tool for defining interesting binary operations on a small set is an *operation table*.

Example 2.2. Let $*$ be the operation on the set $S = \{a, b, c, d\}$ where $i * j$ is the entry in the row indexed by i and the column indexed by j in the following table.

$*$	a	b	c	d
a	a	a	b	c
b	b	b	c	d
c	d	b	c	c
d	b	a	c	d

An operation $*$ on a set S is

i). *commutative* if $a * b = b * a$ for all $a, b \in S$, and

ii). *associative* if $a * (b * c) = (a * b) * c$ for all $a, b, c \in S$.

If an operation $*$ is associative then we can write $a * b * c$ without ambiguity.

Problem 2.3. Is the operation in Example 2.2 commuative or associative? What does the operation table of a commutative operation look like?

Thes properties are usually quite easy to prove, but we will have to do it.

Example 2.3. Let F be the set of all real functions functions $f : \mathbb{R} \to \mathbb{R}$. The operations $+, \cdot, -$ and $\circ$ defined on F by

$$[f+g](r) = f(r) + f(g) \qquad \text{and} \qquad [f \cdot g](r) = f(r) \cdot g(r)$$
$$[f-g](r) = f(r) - g(r) \qquad \text{and} \qquad [f \circ g](r) = f(g(r))$$

are binary operations on F.

The following is Theorem 2.13 of the text.

Problem 2.4. Show that the operation $\circ$ of composition on the set $F(S)$ of functions from S to S is associative.

Defining a binary operation on a set is not always as easy as the examples we have seen. The text finishes this section with a bunch of examples of poorly defined operations. Please take a look at these. If you make any of these mistakes, then I will kill[†] you.

We finish with a non-trivial definition of an operation.

Example 2.4. Recall from Example 0.2 the set $\mathbb{Z}_7$ of equivalence classes mod the modulo relation $\equiv_7$. Let the binary operation $+$ on $\mathbb{Z}_7$ be defined by $\overline{a}+\overline{b} = \overline{a+b}$. Now, is it clear that $\overline{3}+\overline{0} = \overline{3}+\overline{7}$?

If this isn't true, then our definition does not make sense. Remember that that in $\mathbb{Z}_7$ we have that $\overline{0} = \overline{7}$, so these better be the same. We have to check this. We have to check that this operation is *well defined*. Here is how we do it. Pay attention to this! You will do this 100 times over the course of this class.

Let $a' \in \overline{a}$ and $b' \in \overline{b}$. If we can show that $a'+b' \in \overline{a+b}$, then $\overline{a'+b'} = \overline{a+b}$, and so

$$\overline{a'} + \overline{b'} = \overline{a'+b'} = \overline{a+b} = \overline{a} + \overline{b}$$

as needed.

Well $a' \in \overline{a}$ means that $7|a-a'$, and $b' \in \overline{b}$ means that $7|b-b'$. So

$$7|(a-a') + (b-b') = (a+b) - (b+b')$$

which means that $a'+b' \in \overline{a+b}$.

Problems from the Text

Section 2: 5, 6, 9, 13, 14-16, 17, 23, 24, 27, 28, 31, 37

[†]This isn't a footnote, it is the knife that I'll kill you with.

3 Isomorphic Binary Structures

A *binary (algebraic) structure* $\langle B, *\rangle$ consists of a set B and an binary operation $*$ on B. An isomorphism $\phi : \langle B, *\rangle \to \langle B', *'\rangle$ of binary structures is a bijection $\phi : B \to B'$ that satisfies the *homomorphism property*: for all $a, b \in B$ we have

$$\phi(a * b) = \phi(a) *' \phi(b).$$

This is really just a renaming of the elements of B, and the homomorphism property means that it doesn't matter if we do the operation before the renaming, or after. We also say that ϕ *commutes with the operations* to mean that it satisfies the homomorphism property.

If there is an isomorphism $\phi : \langle B, *\rangle \to \langle B', *'\rangle$ then $\langle B, *\rangle$ and $\langle B', *'\rangle$ are *isomorphic*, denoted $\langle B, *\rangle \cong \langle B', *'\rangle$

A map $\phi : B \to B'$ that is not necessarily a bijection and satisfies the homomorphism property is a *homomorphism*.

Example 3.1. Now that we know what isomorphism is, we show that $\langle \mathbb{Z}_n, +\rangle$ defined as in Example 2.4 and the nth roots of unity $\langle U_n, \cdot\rangle$ under complex multiplication, are isomorphic binary structures.

Indeed we show that $\phi : \mathbb{Z}_n \to U_n : \overline{a} \mapsto \zeta_n^a$ is an isomorphism. First we have to check that it is a bijection, but as

$$\mathbb{Z}_n = \{\overline{0}, \overline{1}, \ldots, \overline{n-1}\} \qquad \text{and} \qquad U_n = \{\zeta_n^0, \zeta_n^1, \ldots, \zeta_n^{n-1}\}$$

this is clear from the definition of ϕ. So we only have to check that ϕ *commutes with the operations*: $\phi(\overline{a} + \overline{b}) = \phi(\overline{a}) \cdot \phi(\overline{b})$. Well

$$\phi(\overline{a} + \overline{b}) = \phi(\overline{a+b}) = \zeta_n^{a+b} = \zeta_n^a \cdot \zeta_n^b = \phi(\overline{a}) \cdot \phi(\overline{b}).$$

The following two problems are examples in the text. Try them first, and if you have trouble look in the text. Remember, you have to find a bijection between the sets, show that it is a bijection (injective and surjective), and then show that it commutes with the operations.

Problem 3.1. Show that $\langle \mathbb{Z}, +\rangle$ and $\langle 2\mathbb{Z}, +\rangle$ are isomorphic.

Problem 3.2. Show that the binary stuctures $\langle \mathbb{R}, +\rangle$ and $\langle \mathbb{R}^+, \cdot\rangle$, where $+$ and $\cdot$ are the normal addition and multiplication, are isomorphic. (If finding a bijection here is tricky, here is a hint: how do you usually convert multiplication to addition? Logs, right? Reverse it. Think about exponents.)

To show that two structures are not isomorphic is often much quicker. If the sets have different cardinalities, you are done. Check this first. If they are the same cardinality, usually you have to find some element with a special property in one that does not exist in the other, or an equation that holds in one structure but not the other.

Problem 3.3. Show that $\langle \mathbb{Z}, + \rangle$ and $\langle \mathbb{R}, \cdot \rangle$ are not isomorphic.

Problem 3.4. Show that $\langle \mathbb{R}, + \rangle$ and $\langle \mathbb{R}, \cdot \rangle$ are not isomorphic.

Problem 3.5. Show that $\langle \mathbb{R}, \cdot \rangle$ and $\langle \mathbb{C}, \cdot \rangle$ are not isomorphic.

A *structural property* is a property that is preserved under isomorphism of binary structures. Cardinality is a structural propery, so showing that two structures have different cardinalities means that they are not isomorphic.

Definition 3.2. In a binary structure $\langle B, * \rangle$, an element e is an *identity element* if $e * b = b * e = b$ for all $b \in B$.

Fact 3.3. *A binary structure can have at most one identity element.*

Proof. Assume that e and e' are both identity elements, then

$$e = e * e' = e'.$$

In the first equality we are using that e is an identity, in the second, we are using that e' is. □

Fact 3.4. *The existence of an identity element in a binary strcture is a structural property.*

Proof. Let $\langle B, * \rangle$ have identity e, and let $\phi : \langle B, * \rangle \to \langle B', *' \rangle$ be an isomorphism. We show that $\phi(e)$ is an identity of $\langle B', *' \rangle$. Indeed, for any $b' \in B'$ there is some $b \in B$ such that $\phi(b) = b'$. So

$$b' *' \phi(e) = \phi(b) *' \phi(e) = \phi(b * e) = \phi(b) = b'$$

and with a similar argument we get that $b' = \phi(e) *' b'$. Thus $\phi(e)$ is the identity. □

You can do the following with a very similar proof, and then use it to give another proof to Problem 3.4.

Problem 3.6. An element z of $\langle B, * \rangle$ is *absorbing* if $z * b = z = b * z$ for all $b \in B$. Show that the existence of an absorbing element is a structural property.

Definition 3.5. A structure $\langle B, * \rangle$ is *commutative* if its operation $*$ is commutative.

Problem 3.7. Show that *commutativity* is a structural property for binary structures.

Problems from the Text

Section 3: 2-10, 16, 19, 21-22, 23, 26, 27, 31, 32, 34

4 Groups

A group is just a binary structure with a few nice structural properties. It is the main binary structure that we will be playing with for the rest of the course.

Recall that for a binary operation $*$ on a set G,

i). $*$ is *associative* if $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$,

ii). an element $e \in G$ is an *identity* if $e * a = a * e = a$ for all $a \in G$,

iii). an *inverse* of an element $a \in G$ is an element $a' \in G$ such that $a * a' = e = a' * a$.

Definition 4.1. A *group* $G = \langle G, * \rangle$ consists of a set G and an associative binary operation $* : G \times G \to G$ such that there is an identity element e and every element of G has an inverse. If $*$ is commuative, then G is an *abelian* group. The *order* of the group is $|G|$.

Note

Actually, Question 4.38 of the text asks you to show that if $\langle G, * \rangle$ is a binary structure with an associative operation $*$ such that there is a *left identity* e– an element such that $e * a = a$ for all a– and every $a \in G$ has a *left inverse* a' such that $a' * a = e$, then e is an identity, and the left inverse of a is an inverse. So G is a group. While one can prove this in a couple of lines, they are a bit tricky to come up with. The result is useful though.

Problem 4.1. Which of the following binary structures are groups? For those that are not, explain why they aren't. Which of the groups are abelian?

i). $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ under $+$, and under $\cdot$.

ii). $\mathbb{Z}^*, \mathbb{Q}^*, \mathbb{R}^*$ and $\mathbb{C}^*$ under $+$, and under $\cdot$.

iii). $\mathbb{Z}^+, \mathbb{Q}^+, \mathbb{R}^+$ under $+$, and under $\cdot$.

iv). The set $M_n(\mathbb{R})$ of all $n \times n$ real matrices under $+$ and under matrix multiplication $\times$.

v). The subset $GL_n(\mathbb{R})$ of invertible matrices of $M_n(\mathbb{R})$ under $+$ and $\times$.

vi). The structure $\langle \mathbb{R}, \exp \rangle$ where exp is the operation $\exp : \mathbb{R} \times \mathbb{R} \to \mathbb{R} : (a, b) \mapsto a^b$.

From your answers here, we now define shorter names for several groups

$$\mathbb{Z} := \langle \mathbb{Z}, + \rangle \qquad \mathbb{Q} := \langle \mathbb{Q}, + \rangle \qquad \mathbb{R} := \langle \mathbb{R}, + \rangle \qquad \mathbb{C} := \langle \mathbb{C}, + \rangle$$

$$\mathbb{Q}^* := \langle \mathbb{Q}^*, \cdot \rangle \qquad \mathbb{R}^* := \langle \mathbb{R}^*, \cdot \rangle \qquad \mathbb{C}^* := \langle \mathbb{C}^*, \cdot \rangle$$

We refer to these groups all the time. There are several more that we need in our stable of usual examples.

Example 4.2. For any integer n, the set $\mathbb{Z}_n$ is a group under modular addition $+_n$. Indeed, the identity is $0 = \overline{0}$ and the inverse of $\overline{a}$ is $-\overline{a}$. We will see this group a lot. Let's drop the bars and write it as $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$. We will write $+_n$ as $+$. We showed that $\mathbb{Z}_n$ isomorphic to the set U_n of nth roots of unity under mutliplication, so this is a group too.

Problems 4.11 - 4.18 of the text ask you to determine, for several subsets of the set of $n \times n$ real matrices, if the set is a group under matrix addition or multiplication. Notice that both of these operations are associative on the set $M_n(R)$. As this is the same operation on any subset, it 'inherits' associativity. So you do not have to check this each time. This 'inheritence' of properties will be useful in proofs. Commutativity is also inherited.

The existence of inverse and identity are not inherited by subsets. For these problems you have to check that the set contains the identity, and is closed under the operation and under taking inverses.

Some of these problems:

Problem 4.2. The set $M_n(\mathbb{R})$ of all $n \times n$ real matrices is a group under matrix addition and but not under matrix multiplication. Are the following subsets of $M_n(\mathbb{R})$ groups under matrix addition or matrix multiplication?

- The set of all diagonal matrices in $M_n(\mathbb{R})$.
- The set of diagonal matrices with no 0 on the diagonal.
- The upper triangular matrices.
- The set $GL(n, \mathbb{R})$ of invertible matrices in $M_n(\mathbb{R})$.
- The set $SL(n, \mathbb{R})$ of matrices of determinant 1.

A note on notation

We now usually drop the notation $*$, and use $\cdot$ for most groups; we almost always use $\cdot$ when we are making general statements about groups. When we do this, we use a lot of notation that is common to use with multiplication:

- The identity element is written as e or 1.
- The inverse of a is written as a^{-1}.
- We often drop the $\cdot$ and write ab for $a \cdot b$.

- We use the abreviation $\Pi_{i=1}^{n} a_i$ for $a_1 a_2 \ldots a_n$. As group operations are associative, this makes sense.
- We write a^n for $\Pi_{i=1}^{n} a$ and a^{-n} for $\Pi_{i=1}^{n} a^{-1}$. As this notation seems to require, we let a^0 be 1.

To emphasise the fact that we are using this notation we might refer to G as a *multiplicative group*. But the things that we prove for such groups will still hold for the group with another symbol as the operation.

Occasionally, especially when the group one, like $\mathbb{R}$ or $\mathbb{Z}_n$, in which we are accostomed to use $+$, we will write the group *additively*, using $+$ for the operation. When we do so, the group will always be abelian, the identity is writen e or 0, the inverse of a as $-a$, and we use $\Sigma_{i=1}^{n} a_i$ for $a_1 + a_2 + \cdots + a_n$.

Characterising the finite abelian groups

The following problems start a task that will be one of our main tasks this semester– describing all the finite abelian groups.

Problem 4.3. How many groups are there with two elements?

Say we have a group G with two elements. It must have an identity, so lets say $G = \{1, a\}$. We can write out the operation table

$\cdot$	1	a
1	1	a
a	a	

and there are only two ways to finish it. Oh, actually, as a must have an inverse, $aa = 1$; there is only one way to finish it. It follows that all groups with two vertices are isomorphic.

Problem 4.4. How many groups are there with three elements? That is, how many ways can we finish of the following table.

$\cdot$	e	a	b
e	e	a	b
a	a		
b	b		

It looks like we can finish it off in 3^4 ways. But these won't all be groups. Lets look at some group properties that might help us finish this problem.

Elementary Group Properties

A useful group property is cancellation. In $\mathbb{R}$, if $a + b = a + c$, then we can say $b = c$. We cannot cancel like this with every operation. Certainly $0 \cdot 2 = 0 \cdot 3$ is true, but $2 = 3$ is not. However $\langle \mathbb{R}, \cdot \rangle$ is not a group– 0 doesn't have an inverse.

An operation $\cdot$ on a set S has *left cancellation* if $s \cdot t = s \cdot t' \Rightarrow t = t'$ and has *right cancellation* if $t \cdot s = t' \cdot s \Rightarrow t = t'$.

Using the fact that group operations have inverses, you should be able to prove the following.

Theorem 4.3. *Any group operation has left and right cancellation.*

Proof. See text. □

This gives the following.

Corollary 4.4. *The equation $ax = b$ has a unique solution in a group. So does $ya = b$.*

This helps us a bit with Problem 4.4, right. The elements in the a row are $ae = a, aa$, and ab. But the above theorem says that none of them are the same. So the last two elements in that row are a, b or b, a. The following, Problem 4.31 of the text, lets us eliminate one of these choices.

Problem 4.5. Prove that a group has only one idempotent element: x such that $xx = x$.

Great, I think now you can finish off Problem 4.4 and show that, up to isomorphism, there is only one group of order 3. Problem 4.20 of the text asks you to find all groups of order 4.

Problems from the Text

Section 4: 1, 3, 6, 9, 10, 11-18, 19, 20, 25, 26, 27, 29, 31, 35, 41

5 Subgroups

If a subset $H \subset G$ of the elements of group $G = \langle G, \cdot \rangle$ is closed under the operation of G, then $H = \langle H, \cdot \rangle$ is also a group. It is a *subgroup* of G; we write $H \leq G$ or $G \geq H$. The operation $\cdot$ on H is said to be *induced*. If H is a proper subset of G and a subgroup, then it is a *proper subgroup* of G, written $H < G$ or $G > H$.

As the operation of a group G induced on a subset H inherits associativity, we get the following.

Fact 5.1. *A subset H of G is a subgroup if and only if it contains the identity of G, is closed under the operation of G, and is closed under taking inverses.*

Examples: The additive group $\mathbb{Z}$ is a subgroup of the additive group $\mathbb{Q}$, which in turn, is a subgroup of the additive group $\mathbb{R}$. Every group is a subgroup of itself, and for any group G with identity e, $\{e\}$ is a subgroup of G; it is called the *trivial subgroup*.

We saw in Problem 4.20 that there are only two groups of order 4. We saw that one was isomorphic to the group $\mathbb{Z}_4$ and the other to the following group, which is (for now) called the *Klein* 4*-group*, denoted $V = \mathbb{Z}_2 \times \mathbb{Z}_2$:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

How did you show that they were different? I hope you showed this. One way is to observe that V has subgroups $\{e, a\}, \{e, b\}$ and $\{e, c\}$. An isomorphism $\phi : V \to \mathbb{Z}_4$ takes one of these sets to the set $\{0, 1\}$ in $\mathbb{Z}_4$, but this set is not closed, so does not induce a subgroup.

They have different subgroups, so cannot be isomorphic groups. We will often draw the subgroups of a group, ordered by inclusion, to differentiate the groups. The groups $\mathbb{Z}_4$ and V have the following two different subgraph diagrams.

$$\begin{tikzcd}[column sep=small]
\mathbb{Z}_4 \arrow[d, dash] \\
\{0,2\} \arrow[d, dash] \\
\{0\}
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}[column sep=small]
 & V \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\{e,a\} \arrow[dr, dash] & \{e,b\} \arrow[d, dash] & \{e,c\} \arrow[dl, dash] \\
 & \{e\} &
\end{tikzcd}$$

This is a beautiful picture, but how do we find all the subgroups of a group? What are the subgroups of $\mathbb{Z}$, or of $\mathbb{Z}_{10}$? For $\mathbb{Z}_{10}$ we could check every subset S

of $\mathbb{Z}_{10}$ containing $e = 0$ and check whether it is closed and closed under inverse. But there are 2^{10} such subsets, so this seems like too much work. And we cannot even do this for $\mathbb{Z}$. There are infinitely many subsets. So we have to be smarter.

Well, for $\mathbb{Z}_{10}$, if 1 is in a subgroup $S \subset \mathbb{Z}_{10}$, then $1 + 1$ must be there, and $1 + 1 + 1$. Okay, so there is only one subgroup that contains 1, and it is $\mathbb{Z}_{10}$ itself. This has halved our work already. Let's generalize this idea.

For a group G and an element a of G, the subset of G the set

$$\langle a \rangle := \{a^n \mid n \in \mathbb{Z}\}$$

generated by a.

Theorem 5.2. *Let G be a group and a an element of G, then $\langle a \rangle$ is a subgroup of G. If H is a subgroup of G containing a, then H contains $\langle a \rangle$.*

Proof. The subset $\langle a \rangle$ is certainly closed under the operation, as $a^m \cdot a^n = a^{m+n}$, and is closed under taking inverse as the inverse a^{-m} of any a^m in $\langle a \rangle$ is also in $\langle a \rangle$. It also contains $e = a^0$. On the other and as a subgroup H of G is closed, and closed under inverse, if it contains a, it must contain a^m and a^{-m}. So it contains $\langle a \rangle$. □

The subgroup $\langle a \rangle$ of G is called the *subgroup generated by* a. The *order of* a, as an element of G, is the order of $\langle a \rangle \leq G$.

Problem 5.1. Show that if H and H' are subgroups of G, then $H \cap H'$ is a subgroup of G. If H and H' both contain some $a \in G$ then so does $H \cap H'$. Show that there is a unique smallest subgroup of G that contains a.

As $\langle a \rangle$ is contained in any subgroup containing a, it is the smallest subgroup containing a. Sometimes this is the definition of $\langle a \rangle$. A subgroup H of G is called *cyclic* if $H = \langle a \rangle$ for some a in H, and the element a is called its *generator*.

Example 5.3. For any positive integer $n \geq 2$ the group $\mathbb{Z}_n$ is generated by the element 1, so is cyclic. The group $\mathbb{Z}$ is generated by 1 so is cyclic. The group $\mathbb{Z}_4$ has two different generators: $\langle 1 \rangle = \langle 3 \rangle = \mathbb{Z}_4$, but $\langle 0 \rangle = \{0\}$ and $\langle 2 \rangle = \{0, 2\}$. The group V is not cyclic, as $\langle a \rangle$, $\langle b \rangle$, and $\langle c \rangle$ both have two elements.

Problem 5.2. What are the subgroups of $\mathbb{Z}$? Hint: show that all subgroups are cyclic, generated by an element of minimum absolute value.

Problem 5.3. What are the subgroups of $\mathbb{Z}_{10}$?

Problem 5.4. Is the additive group $\mathbb{R}$, or the multiplicative group $\mathbb{R}^+$, cyclic?

Problems from the Text

Section 5: 1-6, 14-19, 24, 26, 33, 36, 37-38, 39, 42, 43, 46, 51, 52, 57

6 Cyclic Groups

One of our goals will be to describe all finite abelian groups. We will build them all from cyclic groups, so here we look at properties of cyclic groups, and describe them all.

Theorem 6.1. *Every cyclic group is abelian.*

Proof. If a group G is cyclic, we can write it as $\langle a \rangle$ for some element a. So any two elements g and g' can be written as $g = a^m$ and $g' = a^n$ for some m and n. So

$$gg' = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = g'g.$$

□

In a finite cyclic group, we will have that $a^m = a^0 = e$ for some $m \geq 1$. So for any $n = mq + r$ we can write a^n as

$$a^n = (a^m)^q \cdot a^r = 1^q \cdot a^r = a^r.$$

Recall the division algorithm, you would have proved this in a discrete math or number theory class.

Theorem 6.2. *For fixed integer $m \geq 1$ and any integer n, there are unique integers q, and r, with $0 \leq r < m$ such that*

$$n = qm + r.$$

Proof. Fix m. We prove existence, for $n \geq 0$, by induction. If $n < m$, then $(q, r) = (0, n)$ are such q and r. So assume that $n \geq m$. Then where $n' = n - m$ we have, by induction, that $n' = q'm + r'$, and so $n = n' + m = (q' + 1)m + r'$. Now, if $n < 0$, then $-n = q'm + r'$ for some q' and r' with $0 \leq r' < m$. So $n = -q'm - r'$. If $r' = 0$ we are done, otherwise $n = (-q - 1)m + (m - r')$ and $0 < (m - r') < m$.

For uniqueness, assume that $qm + r = n = q'm + r'$. As $r - r' = (q' - q)m$ we have that $m | r - r'$. But they are both between 0 and $m - 1$, so $r = r'$. But then $qm = q'm$ and so $q = q'$. □

I'm going to take it for granted that you can do the following.

Problem 6.1. Find the *quotient* q and the *remainder* r such that $37 = 9q + r$.

Theorem 6.3. *Every subgroup of a cyclic group is cyclic.*

Proof. See text. □

We saw that in the group $\mathbb{Z}$ the cyclic sugroup $\langle n \rangle = n\mathbb{Z}$ is the set of multiples of n. The above theorem tells us that these are the only subgroups of $\mathbb{Z}$.

Corollary 6.4. *The subgroups of $\mathbb{Z}$ under addition are the groups $n\mathbb{Z}$ for positive integers n.*

This gives a real fancy way to define the greatest common divisor $\gcd(m, n)$ of two integers.

Definition 6.5. The gcd, or *greatest common divisor* $\gcd(m, n)$ of integers m and n is the positive generator of the subgroup $H = \{am + bn \mid a, b \in \mathbb{Z}\}$ of $\mathbb{Z}$.

As m and n are in $H = \langle d \rangle = d\mathbb{Z}$, it is clear that d divides m and n. If there were a larger number d' that divided them both, then it would divide everything of the form $am + bn$ and so would contain H. But this would contradict the fact that d generates H.

Theorem 6.6. *Any cyclic group is isomorphic to $\mathbb{Z}$, or to $\mathbb{Z}_m$ for some positive integer m.*

Proof. Let G be a cyclic group with generator a. First, we show that if there is no integer $m \geq 1$ such that $a^m = e$, then the map $\phi : \mathbb{Z} \to G : z \mapsto a^z$ is an isomorphism. It is certainly surjective, as $G = \{a^z \mid z \in \mathbb{Z}\}$. To show that it is injective we show that there are no two integers $h < k$ such that $a^h = a^k$. Indeed, if there were, then we would have $a^{h-k} = e$, contradicting the fact that there is no $m \geq 1$ with $a^m = e$. That this ϕ commutes with the operations is clear:

$$\phi(h + k) = a^{h+k} = a^h \cdot a^k = \phi(h) \cdot \phi(k).$$

So we may assume that there is some m, (and so some minimum $m \geq 1$), such that $a^m = 1$. We now show that $\phi : \mathbb{Z}_m \to G : z \mapsto a^z$ is an isomorphism. That it is a injection is clear, as if $a^h = a^k$ for $0 \leq k < h \leq m$ then $a^{h-k} = e$ and $0 < h - k < m$, contradicting our choice of m. As the sets have the same cardinality, this shows that ϕ is a bijection. The proof that it commutes with the operations is as above. $\square$

Subgroups of Finite Cyclic Groups

We saw exactly what the cycle groups are upto isomorphism. They are the groups $\mathbb{Z}_n$ for integers n, and $\mathbb{Z}$. We also saw exactly what the subgroups of $\mathbb{Z}$ are: the groups $n\mathbb{Z}$ for integers n. Now we look at what the subgroups of the cyclic group of order n are.

The cyclic group of order n is $\mathbb{Z}_n$. But remember this is only unique upto isomorphism. It is also U_n. As it is generated by a single element, we often write it as $\langle a \rangle$ for some generic generator a.

Let's start with $\mathbb{Z}_{12}$ and find all of its subgroups. We know from Theorem 6.3 that they are all cyclic. So they are all of the form $\langle s \rangle$ for $s \in \mathbb{Z}_{12}$.

Problem 6.2. List the elements of $\langle s \rangle$ for each $s \in \mathbb{Z}_{12}$. Do any elements give the same subgroups? Which elements are generators of $\mathbb{Z}_{12}$? Can you write a formula that predicts, for a given s, how big the subgroup it generates is? Are there different subgroups of the same size? Draw the subgroup diagram for $\mathbb{Z}_{12}$.

Theorem 6.7. *The subgroup $\langle s \rangle$ of $\mathbb{Z}_n$ for $s \in \mathbb{Z}_n$ has order n/d where $d = \gcd(s, n)$; it is the set $\langle d \rangle = \{zd \mid z \in [n/d]\}$. In particular $\langle s \rangle = \langle t \rangle$ if $\gcd(s, n) = \gcd(t, n)$, and s generates $\mathbb{Z}_n$ if and only if $\gcd(s, n) = 1$.*

Proof. Let $d = \gcd(s, n)$. By definition $\langle s \rangle = \{zs \in \mathbb{Z}_n \mid z \in \mathbb{Z}\}$, but as n divides $(n/d)s$ we have that

$$is \equiv_n (i + n/d)s \equiv_n (i + 2n/d)s \equiv_n \ldots,$$

and so $\langle s \rangle = \{zs \in \mathbb{Z}_n \mid z \in [n/d]\}$, which has at most n/d elements.

On the other hand, as $d = \gcd(s, n)$, we have that $d = as + bn$ for some integers a and b. So $d \equiv_n as + bn \equiv_n as$, which means $d \in \langle s \rangle$ and so $\langle d \rangle \subset \langle s \rangle$. But $d|n$ so $\langle d \rangle = \{d, 2d, \ldots, n\}$ has exactly n/d elements, and so $\langle d \rangle = \langle s \rangle$.

□

Problem 6.3. The above theorem is written in multiplicative form as Theorem 6.14 of the text. Without looking at the text try to rewrite in multiplicative form: 'Let $G = \langle a \rangle$ be a cyclic group of order n. The subgroup $\langle a^s \rangle$ generated by an element $a^s \in G$...'

Now that we have some tools, we can do Problem 6.2 more quickly. Lets try it with another cyclic group.

Problem 6.4. Draw the subgroup diagram of $\mathbb{Z}_{18}$.

Solution: We get one subgroup generated by each divisor d of 18. The divisor 1 generates $\mathbb{Z}_{18}$ (as do all the elements that are relatively prime to 18 : $\{1, 5, 7, 11, 13, 17\}$). The divisor 2 generates $\langle 2 \rangle = \{2, 4, 6, 8, 10, 12, 14, 16, 0\}$, 3 generates $\{3, 6, 9, 12, 15, 0\}$, 6 generates $\langle 6 \rangle = \{6, 12, 0\}$, which is a subgroup of both $\langle 2 \rangle$ and $\langle 3 \rangle$ because $2, 3|6$, etc. Now draw that diagram!

Problems from the Text

Section 6: 1, 6, 11, **14-15**, 18, 21, 28, 30, 31, **32**, **33-37**, 43, 44, 46, **48**, 49, **51**, 55

7 Generating Sets and Cayley Digraphs

We showed that the cyclic subgroup $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$ of a group G is the smallest subgroup of G that contains a. What should the subgroup $\langle a, b \rangle$ of G generated by a and b be? Of course, it should contain every thing that we can get from a and b by products and inverses. It should contain such elements as:

$$a, b, ab, ab^2, a, b^{-2}, a^m b^n, aba, a^{m_1} b^{n_1} a^{m_2} b^{n_2}.$$

We can say this rigorously, but instead, we are going to define $\langle a, b \rangle$ as the smallest subgroup of G containing a and b. But is there a unique smallest subgroup containing a and b? We first have to prove this before making this definition.

Theorem 7.1. *For a family $\{H_i \mid i \in I\}$ of subgroups H_i of a group G, the set $H = \cap_{i \in I} H_i$ is a subgroup of G.*

Proof. Let a and b be in H. As H_i is a subgroup, all of ab, e, and a^{-1} are in H_i for all i. So they are all in H. That is, H is closed, contains the identity, and is closed under inverses. So H is a subgroup of G. □

Corollary 7.2. *For any subset S of a group G, there is a unique subgroup $\langle S \rangle$ of G containing all elements of S and contained if every subgroup of G that contains S.*

Proof. As G contains S there are subgroups that contain S. Let $\{H_i \mid i \in I\}$ be the family of such subgroups of G. By the theorem the set $\langle S \rangle := \cap_{i \in I} H_i$ is a group, which contains S, and is contained in all H_i. If there is another such group H, then $H \subset \langle S \rangle$ and $\langle S \rangle \subset H$, and so $H = \langle S \rangle$. □

With this we have the following definition.

Definition 7.3. For a subset S of a group of G call $\langle S \rangle$ the **the subgroup of G generated by S**. If $\langle S \rangle = G$ then S *generates* G, and if S is finite, then G is *finitely generated*.

If a is in a set S that generates G, we call a **a generator** of G. There is a possibility of confusion when saying this, as it might sound like one is saying that G is cyclic. One should be clear about this when using this terminology.

Now, we should show that this agrees with our other intuitive notion of $\langle S \rangle$

Theorem 7.4. *Let G be a group and S be a set of elements of G. The subgroup $H = \langle S \rangle$ of G consists of those elements of G that can be written as a finite product of integral powers of element of S.*

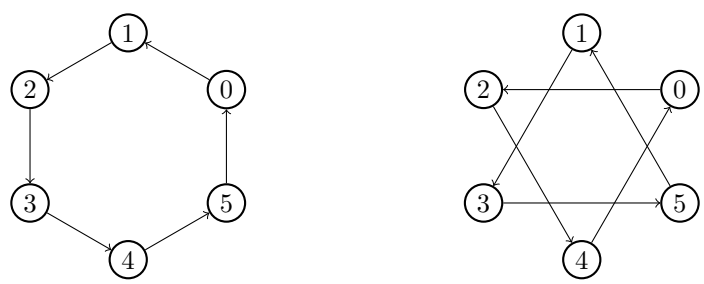
Proof. It is clear that the $S \subset K \subset H$, so it is enough to show that K is a group. A product of two finite products is still finite, so K is close under products. For any $a \in S$ it contains $a^0 = e$, so K contains the identity. And for some element $\prod_{i=1}^{n} s_i^{d_i}$ of K the inverse $\prod_{I=n}^{1} s_i^{-d_i}$ is also in K, so K is a group. $\square$

Cayley Digraphs

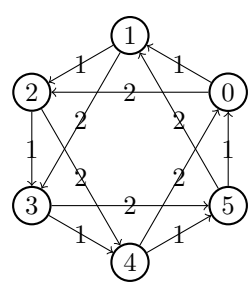
There is a nice way to visualise the operations of a group in terms of generators.

The *Cayley digraph* of a group G and a set S of generators of G is the directed graph $\Gamma(G, S)$ with vertex set G and in which (u, v) is an arc if there is some s in S such that $us = v$.

Example 7.5. Here are $\Gamma(\mathbb{Z}_6, \{1\})$ and $\Gamma(\mathbb{Z}_6, \{2\})$.



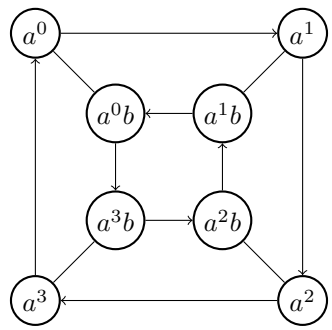
Sometimes we view $\Gamma(G, S)$ as an arc-coloured digraph, saying the arc (u, v) has colour s if $us = v$. Here is $\Gamma(\mathbb{Z}_6, \{1, 2\})$.



Observe that a generating set of a group need not be a minimal set that generates the group. Usually if (u, v) and (v, u) are both arcs, we simply draw them together as an edge without direction.

Problem 7.1. Draw $\Gamma(\mathbb{Z}_6, \{2, 3\})$.

The Cayley graph of a group plays a similar role to the multiplication table– it tells you exactly how to multiply two elements.



Problem 7.2. Where $\Gamma(G, \{a, b\})$ is drawn above, (symmetric edges are b-colouring in both directions), compute $(a^2b)(ab)$.

A Cayley digraph has three important properties:

i). It is *vertex transitive*: for any two vertices u and v there is a bijection $\phi : G \to G$ (called an automorphism) such that uv is a c coloured arc if and only if $\phi(u)\phi(v)$ is a c-coloured arc.

ii). The vertex e (so every vertex,) has exactly one arc of each colour leaving it.

iii). The graph is connected: for every pair u and v of vertices, there is a sequence of arcs walking from u to v.

Problem 7.3. Prove that a Cayley graph has the above three properties.

One can show that if a digraph has an edge colouring that satisfies these three properties, then it is the Cayley graph of a group. Use this in Exercise 7.19. The group of size 8 whose Cayley digraph is given above answers this question in the case of $n = 4$.

Problem 7.4. One can consider the Cayley digraph $\Gamma(G, S)$ even when the subset S does not generate G. What property does the digraph have if and only if S is a generator? What conditions on S would you need so that $\Gamma(G, S)$ is a symmetric graph?

Problems from the Text

Section 7: 3, 7, **10**, **11**, 12, 14, 16,19

8 Groups of Permutations

We saw a binary structure $\langle F(G), \circ \rangle$ of functions $f : G \to G$ on a set G with the operation composition. With respect to composition, the identity element is the identity map id, and any element has an inverse if it is a bijection. So the set of bijections $\sigma : G \to G$ is a group under composition. We have another name for bijections of G to G– they are *permutations* of G.

Example 8.1. There are 3! permutations of the set $[3] = \{1, 2, 3\}$. Possibly you've seen the permutations of a set as its orderings, so the permutations of $[3] = \{1, 2, 3\}$ are (written compactly)

$$123, 132, 213, 231, 312, 321.$$

We are looking at them as functions now so the last of these is the function σ that has

$$\sigma(1) = 3 \qquad \text{and} \qquad \sigma(2) = 2 \qquad \text{and} \qquad \sigma(3) = 1.$$

There are a couple of different useful notations for permutations. We will denote the above sigma as $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{smallmatrix}\right)$, the first row just reminding us what is being permuted.

Written this way, we usually think of the permutation as an object, and write composition as multiplication. If $\tau = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{smallmatrix}\right)$, then we write

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix},$$

so that, for example, $(\sigma\tau)(2) = \sigma(\tau(2)) = \sigma(1) = 3$.

Let S_A be the group of permutations of a set A; and for an integer $n \geq 1$ let $S_n = S_{[n]}$. The group S_n is the *symmetric group on n elements.*

Problem 8.1. I might misspeak and call S_n the *symmetric group of order n*, but its order is NOT n. What is the order of S_n?

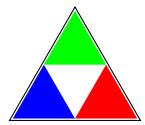
The following should be easy.

Problem 8.2. Show that if $|A| = n$ then $S_A \cong S_n$.

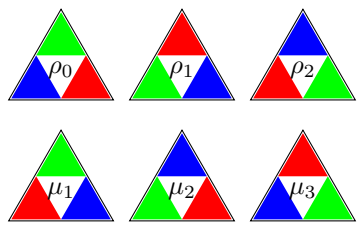
Do not mistake 'symmetric' for 'abelian'! In fact, we have $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{smallmatrix}\right)$ while $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{smallmatrix}\right)$, so S_3 is not abelian. We saw that all groups of order $2, 3$ and 4 are abelian, and we will see that $\mathbb{Z}_5$ is the only group of order 5, so S_3 is a smallest non-abelian group.

There is another setting in which the group S_3 arises. It is isomorphic to the group D_3 of symmetries of an equilateral triangle. This is an important group, and it does not hurt to take some time to become familiar with it.

Example 8.2. To explain the group D_3, think of an equilateral triangle whose corners are coloured with the colours $1, 2$ and 3. It looks like this:



Pretend that there is a little depression in your desk that you can fit it in to. You can fit it in, in six different ways:



Each placement gives us a permutation of the colours, we have given them names in the figure. The identity permuation is ρ_0. You can see that the subgroup generated by the 'rotation' ρ_1 consists of three rotations. Adding in any 'flip' μ_i, we generate the whole group.

Problem 8.3. Write each element of D_3 as a permutation of the set $\{1, 2, 3\}$.

Problem 8.4. Draw the Cayley graph $\Gamma(D_3, \{\rho_1, \mu_1\})$. Compare this to the graph $\Gamma(\mathbb{Z}_6, \{2, 3\}$ that you drew in the last section.

More generally the nth *dihedral goup* D_n is the group of symmetries of the regular n-gon.

Example 8.3. The 4th-dihedral group D_4, also called the octic group, has a similar presentation- put those numbers $\{1, 2, 3, 4\}$ on the corners. There are eight possible permutations. The identity permuation is ρ_0, and those that we get from ρ_0 by succesive clockwise quarter turns are the *rotations*

$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \qquad \rho_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \qquad \rho_3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}.$$

Those that we get by flipping around a median drawn parallel to an edge are the *medial flips*

$$\mu_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \qquad \mu_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

and those that we get by flipping around a diagonal line are the *diagonal flips*

$$\delta_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \qquad \delta_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}.$$

Teacher

Turns out that μ is the greek version of m, not u. Capital μ is M.

We can show that the subgroup diagram of D_4 is as in Figure 1.

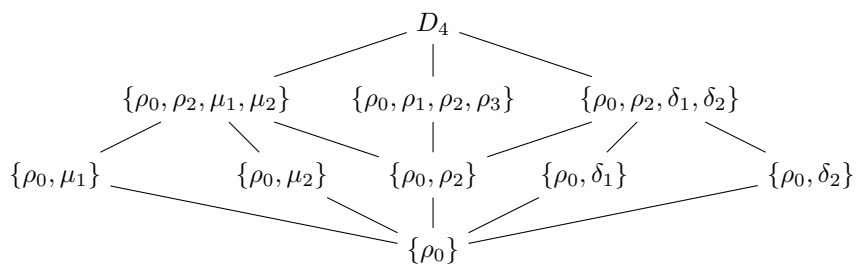


Figure 1: Subgroup diagram of D_4

Cayley's Theorem*

The dihedral group D_4 is a subgroup of the symmetric group S_4. Its elements are permutations of [4]. It is thus a *group of permutations*. It turns out that every group is.

We have observed that each row of the operation table of a group is a permutation of the elements of the group. From this we get an injection from G to the set S_G of permutations on G, and can show that it has the homomorphism property. It follows that G is isomorphic to a subgroup of S_G, and so every group is the subgroup of some symmetric group, so is a group of permutations. This is Cayley's theorem, and it was a little quick. Let's take it a bit slower.

Recall that for groups G and G' a function $\phi : G \to G'$ is a *homomorphism* if it satisfies the homomorphism property:

$$\phi(ab) = \phi(a)\phi(b).$$

Lemma 8.4. *Let $\phi : G \to G'$ be an injective homomorphism. Then $\phi[G]$ is a subgroup of G' and $\phi : G \to \phi[G]$ is an isomorphism.*

Proof. See text. □

Theorem 8.5 (Cayley's Theorem)**.** *Every group is isomorphic to a group of permutations.*

Proof. By the lemma, it is enough to show that the map $G \to S_G$, taking an element a to its row of the operation table, suggested at the start of this subsection is an injective homomorphism. The map is $\phi : G \to S_G : a \mapsto \lambda_a$ where λ_a is the permutation $\lambda_a(x) = ax$ that multiplies an element x by a on the left.

*It's his theorem, but anyone can use it.

We know that ϕ is injective because no element occurs twice in the same column of the opertation table: For $a \neq b \in G$ and some $x \in G$, we have by cancellation that $ax \neq bx$ and so

$$\phi(a)(x) = \lambda_a(x) = ax \neq bx = \lambda_b(x) = \phi(b)(x).$$

So $\phi(a) \neq \phi(b)$, showing that ϕ is injective.

It is a homomorphism as for a, b and all x we have

$$\lambda_{ab}(x) = abx = \lambda_a(bx) = \lambda_a(\lambda_b(x)) = (\lambda_a\lambda_b)(x),$$

and so $\phi(ab) = \lambda_{ab} = \lambda_a\lambda_b = \phi(a)\phi(b)$. $\square$

Problems from the Text

Section 8: 1, 2, **8,11**, 14, 16, 18, **35**, 36, **38**, 39, 43, 45, **46**, 48, 51

9 Orbits, Cycles, and the Alternating Groups

The exercises of the last section introduced the *orbit*

$$\mathcal{O}_{a,\sigma} := \{\sigma^n(a) \mid n \in \mathbb{Z}\}$$

of an element a of a set A under a permutation $\sigma \in S_A$.

Problem 9.1. Show that the relation $\sim$ defined by $a \sim b$ if $b \in \mathcal{O}_{a,\sigma}$, is an equivalence relation on A.

The equivalence classes of the relation in the previous problem are the *orbits of* σ in A. As this is an equivalence relation, the orbits of σ partition A.

Problem 9.2. What are the orbits of the identity permutation on a set A?

Problem 9.3. What are the orbits of $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}$ in S_4?

Solution

The orbits are $\mathcal{O}_{1,\sigma} = \{\sigma^0(1), \sigma^1(1), \sigma^2(1)\} = \{1, 2, 3\} = \mathcal{O}_{2,\sigma} = \mathcal{O}_{3,\sigma}$ and $\mathcal{O}_{4,\sigma} = \{4\}$.

Two different permutations in S_n can have the same orbits, but a permutation also gives an ordering of the elements of an orbit. Indeed in the above problem, the orbit $\mathcal{O}_{1,\sigma} = \{1, 2, 3\}$ has a natural ordering $(\sigma^0(1), \sigma^1(1), \sigma^2(1)) = (1, 3, 2)$. This is called a *cycle* of σ. Every orbit $\mathcal{O}$ of σ yields a cycle of σ by picking an element $a \in \mathcal{O}$ and ordering the orbit as $(a, \sigma^1(a), \sigma^2(a), \ldots, \sigma^{d-1}(a))$ where $d = |\mathcal{O}|$.

Example 9.1. The permutation $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}$ in S_4 has cycles

$$(1, 3, 2) \qquad \text{and} \qquad (4).$$

These cycles have order 3 and 1 so σ has order 3. We can recover σ from its cycles, the first tells us that

$$1 \xrightarrow{\sigma} 3 \xrightarrow{\sigma} 2 \xrightarrow{\sigma} 1 \qquad \text{and} \qquad 4 \xrightarrow{\sigma} 4.$$

Problem 9.4. Show that the permutation with cycles $(3, 2, 1)$ and (4) is the same as the permutation with cycles $(1, 3, 2)$ and (4). Show that these are different from the permutation with cycles $(1, 2, 3)$ and (4). When do two orderings of the parts of a partition give the same permutation?

A cycle of a permutation is *trivial* if it contains only one element. The *cycle notation* of a permutation consists of a concatenation of its non-trivial cycles.

Example 9.2. The permutation $\sigma = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 1 & 4 & 6 & 5 \end{smallmatrix}\right)$ has cycle notation

$$\sigma = (13)(56).$$

Problem 9.5. Give cycle notation for the following permutations in S_6: $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 2 & 4 & 6 & 5 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 3 & 2 & 1 \end{smallmatrix}\right)$.

You observed above that (134) is the same cycle as the cycle (341) that we get from it by permuting its elements cyclically. (Maybe this is why it is called a cycle.) Maybe you also notices that it doesn't matter if we write it as (13)(56) or (56)(13). Keep this in mind. Apart from the obvious benefit of saving a lot of writing, this notation is useful in composing permutations.

Example 9.3. To multiply the two permutations (13452) and (152)(46) in S_6 we write

$$(13452)(152)(46) = (1 \qquad),$$

and then watch where 1 goes as we proceed through the cycles (13452)(152)(46) from right to left. It is fixed by (46), and goes

$$1 \xrightarrow{(46)} 1 \xrightarrow{(152)} 5 \xrightarrow{(13452)} 2.$$

So we write

$$(13452)(152)(46) = (12 \qquad),$$

and then compute where 2 goes. Finish this off!

Problem 9.6. Compute (152)(46)(13452). Is it the same as (13452)(152)(46)?

Oh no! When can we 'commute cycles?

Problem 9.7. Show that if two cycles $(a_1a_2\dots a_m)$ and $(b_1b_2\dots b_n)$ are disjoint then

$$(a_1a_2\dots a_m)(b_1b_2\dots b_n) = (b_1b_2\dots b_n)(a_1a_2\dots a_m)$$

.

Even and odd permutations

In a permutation, a cycle of length 2 is a *transposition.*

Fact 9.4. *Every permutation in S_n for $n \geq 1$ can be written as a product of transpositions.*

Proof. Let σ be in S_n. If σ is the identity permutation, then because $n \geq 2$ we can write it as $\sigma = (12)(12)$. So we may assume that σ is composed of disjoint cycles of the form $\sigma = (a_1a_2\dots a_m)$. We claim that

$$(a_1a_m)(a_1a_{m-1})\dots(a_1a_3)(a_1a_2) = \sigma.$$

Check this by multiplying it out– $a_1 \xrightarrow{(a_1a_2} a_2$ and then a_2 never appears again, so we have that it begins as $(a_1a_2 \qquad)$. Then we have $a_2 \xrightarrow{(a_1a_2)} a_1 \xrightarrow{(a_1a_3)} a_3$, and is done, so we have $(a_1a_2a_3 \qquad)$. Any a_i goes $a_i \xrightarrow{(a_1a_i)} a_1 \xrightarrow{(a_1a_{i+1})} a_{i+1}$ and is done. This gives the claim. $\square$

Now, you probably noticed that this representation of a permutation as a product of transpositions is not unique. But we can say the following.

Theorem 9.5. *No permutation can be writen both as a product of an even number of transpositions, and an odd number of transpositions.*

Proof. Observe that every permutation σ in $\S_n$, viewed as a permutation of the set of rows of the $n \times n$ identity matrix I_n, can be written as a matrix $M = [m_{ij}]$ of with a 1 in $m_{i\sigma i}$ for all i, and a 0 elsewhere. The determinant of M is 1 or -1 depending on whether we can get it from I_n via an even number of row switches, or an odd number, and this is well defined. A row switch is just a transposition of rows, so a series of row switches to get M from I_n is a representation of M as a product of transpositions. □

This allows us to define the following.

Definition 9.6. A permutation in S_n is *even* or *odd* according to whether it can be expressed as a product of an even, or odd, number of transpositions.

Problem 9.8. Are the following permutations even or odd: id, $(1, 3, 4, 2)$, $(5, 6, 7)$, $(1, 3, 4, 2)(5, 6, 7)$?

Fact 9.7. *For $n \geq 2$, half of the permutations in S_n are even, and half are odd.*

Proof. Let A be the set of odd permutations in S_n and B the set of even ones. We claim that $\lambda : B \to A : \sigma \mapsto (12)\sigma$ is a bijection. Indeed its inverse is $\lambda^{-1} : A \to B : \sigma \mapsto (12)\sigma$, this does it. □

Was that too slick? If you have not seen this before do the following.

Problem 9.9. Show that a map $\phi : A \to B$ from a set A to B is a bijection if and only if it has an inverse.

Theorem 9.8. *For $n \geq 2$, the set A_n of all even permutations in S_n is a subgroup.*

Proof. Prove this! □

The group A_n of even permutations of S_n is called the *alternating group of order n*.

Problems from the Text

Section 9: 2, 4, 9, 10, **13**, **17**, 19, 20-22, **23**, **27**, **29**, 30, **34**

10 Cosets and the Theorem of Lagrange

We saw that for a cyclic subgroup $\mathbb{Z}_n$ of finite order, there was a subgroup of order d for every d dividing n, indeed it was the cyclic subgroup $\langle n/d \rangle$. Moreover we saw that these were the only subgroups of $\mathbb{Z}_n$. So all subgroups have order dividing the order of $\mathbb{Z}_n$. Lagrange showed that his holds not only for cyclic groups. We will see that the order $|H|$ of any subgroup H of a group G divides the order $|G|$ of G. The rough strokes of the proof will be to show that for a subgroup H of G, we can partition G up into 'shadows' of $H-$ subsets derived from H having the same size as H. These 'shadows' are called cosets, and will be important. They are central to most of our deeper structural theorems about groups.

Cosets

Let G be a group and H be a subgroup. For any element $g \in G$ the sets

$$aH = \{ah \mid h \in H\} \qquad \text{and} \qquad Ha = \{ha \mid h \in H\}$$

are *left*, and respectively *right, cosets of* H.

Define a relation $\sim$ on G by $a \sim b$ if $aH = bH$.

Fact 10.1. *The relation $\sim$ on G is an equivalence relation, and so the left cosets of H in G partition G.*

Proof. This is easy. Do it as an exercise. □

There are several useful conditions that equivalent to $aH = bH$.

Fact 10.2. *Let H be a subgroup of G, and a, b be in G. The following are equivalent.*

- $aH = bH$
- *There exists $h \in H$ such that $ah \in bH$.*
- *There exists $h, h' \in H$ such that $ah = bh'$.*
- $b \in aH$
- $b^{-1}a \in H$

Problem 10.1. Prove the above fact.

Example 10.3. We find the left cosets of $H = \langle 5 \rangle = \{0, 5, 10, 15\}$ in $\mathbb{Z}_{20}$. Notice that because $\mathbb{Z}_{20}$ is an additive group, the condition $b^{-1}a \in H$ becomes $a - b \in \langle 5 \rangle$, and the left coset containing 3 is denoted $3 + \langle 5 \rangle$. The left cosets are

- $0 + \langle 5 \rangle = \langle 5 \rangle = \{0, 5, 10, 15\}$
- $1 + \langle 5 \rangle = \{1, 6, 11, 16\}$
- $\ldots$
- $4 + \langle 5 \rangle = \{4, 9, 14, 19\}$

What are the right cosets?

Problem 10.2. Show that if G is abelian, then the left cosets of a subgroup H are the same as the right cosets.

This should not be true if G is not abelian right? Our smallest non-abelian group was the 3-dihedral group $D_3 (= S_3)$ with generators the rotation ρ_1 and the flip μ_1. The left and right cosets of the subgroup $H = \langle \rho_1 \rangle$ are

$$\rho_0 H = \{\rho_0, \rho_1, \rho_2\} = H\rho_0 \qquad \text{and} \qquad \mu_1 H = \{\mu_1, \mu_2, \mu_3\} = H\mu_1,$$

while for the subgroup $K = \langle \mu_1 \rangle = \{\rho_0, \mu_1\}$ we have

$$\rho_0 K = \{\rho_0, \mu_1\} \qquad \rho_1 K = \{\rho_1, \mu_3\} \qquad \rho_2 K = \{\rho_2, \mu_2\}$$

and

$$K\rho_0 = \{\rho_0, \mu_1\} \qquad K\rho_1 = \{\rho_1, \mu_2\} \qquad K\rho_2 = \{\rho_2, \mu_3\}.$$

On page 99 of the text, the operation table of D_3 is coloured– each coset of a subgroup H gets a colour. In the case that left and right cosets are the same, this colouring makes a nice pattern, (if we properly order the elements), and this will lead to some structure that we will make use of later. This will be a nice property that we will love to have in a subgroup H. If the left and right cosets are not the same, this colouring is an unholy mess.

Okay. Back to...

The Theorem of Lagrange

Whenever we took a subgroup H of a group G above, the cosets of H all had the same size as H. This always happens.

Lemma 10.4. *For a subgroup H of G and an element g of G, the map*

$$\phi : H \to gH : h \mapsto gh$$

is a bijection.

Proof. That it is surjective is from the definition $gH = \{gh \mid h \in H\}$, and that it is injective comes from the fact that $x \mapsto g^{-1}x$ is the inverse of ϕ. $\square$

As the left cosets of H in G partition G, where there are r left cosets we have from the lemma that $|G| = r|H|$, so we get the following.

Theorem 10.5 (Theorem of Lagrange)**.** *If H is a subgroup of a finite group G, then $|H|$ divides $|G|$.*

You should be able to prove the following two corollaries pretty quickly from the theorem.

Corollary 10.6. *Every group of prime order is cyclic.*

Corollary 10.7. *The order of an element of a finite group divides the order of the group.*

The *index* $(G : H)$ of H in G is the number $|G|/|H|$ of left cosets of H in G. You prove the following in exercise 10.38. (Note that the proof is just counting if you assume that K is finite, but you cannot assume this.)

Theorem 10.8. *For groups $K \leq H \leq G$ such that the indices $(G : H)$ and $(H : K)$ are finite we have $(G : K) = (G : H)(H : K)$.*

This begs the question, if d divides $|G|$, is there a subgroup of G of order d.

Problems from the Text

Section 10: 6, 8, **12, 15**, **19**, **20-24**, 26, **28-29**, **30-33**, **35**, 37, 38, 40

11 Direct Products and Finitely Generated Abelian Groups

The *(cartesian) product* of a finite number of sets $S_1, \ldots, S_n$ is the set

$$\prod_{i=1}^{n} S_i = \{(s_1, \ldots, s_n) \mid \forall i, s_i \in S_i\},$$

which has $\prod_{i=1}^{n} |S_i|$ elements. When the sets are the sets of elements of groups, there is an easy way to make the product into a group.

For groups $G_1, \ldots, G_n$ *componentwise multiplication* is the operation defined on the set $\prod_{i=1}^{n} G_i$ by

$$(a_1, \ldots, a_n)(b_1, \ldots, b_n) = (a_1 b_1, \ldots, a_n b_n).$$

Theorem 11.1. *For groups $G_1, \ldots, G_n$ the set $\prod_{i=1}^{n} G_i$ is a group under componentwise multiplication.*

Proof. See text. □

This group is called the *direct product* of the groups G_i.

Note

If the groups are written additively, especially if they are abelian, one sometimes writes $\oplus_{i=1}^{n} G_i$ instead of $\prod_{i=1}^{n} G_i$ and calls the group the *direct sum* instead of the direct product.

Example 11.2. The group $\mathbb{Z}_2 \times \mathbb{Z}_2$ has element set

$$\{(0,0), (0,1), (1,0), (1,1)\}.$$

As $\mathbb{Z}_2$ is abelian, and addition is component-wise, $\mathbb{Z}_2 \times \mathbb{Z}_2$ is also abelian:

$$(a,b) + (a',b') = (a+a', b+b') = (a'+a, b'+b) = (a',b') + (a,b).$$

Moreover it is not cyclic as clearly every element has order 1 or 2. So it is the Klein group V. And it feels a much nicer representation of this group than we saw before, its structure seem very clear.

So when can we represent a group as a product of smaller groups? A group is *decomposible* if it is isomorphic to a direct product of two proper subgroups. If it is not, then it is *indecomposible*. In this section we will see exactly a finite abelian group can be decomposed into indecomposible subgroups.

Let us start by looking at a simpler set of groups though: groups that are already a product of finite cyclic groups, and lets see what groups we can build from them.

Though the product of cyclic groups need not be cyclic, as we saw with the Klein group, often we do get a cyclic group from a product.

Example 11.3. The group $G = \mathbb{Z}_2 \times Z_3$ has six elements

$$(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)$$

and is generated by $(1,1)$ as the Cayley digraph $\Gamma_{G,\{(1,1)\}}$ shows

$$(0,0) \to (1,1) \to (0,2) \to (1,0) \to (0,1) \to (1,2) \to (0,0)$$

and so $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$.

On the other hand the group $\mathbb{Z}_3 \times \mathbb{Z}_3$ is not cyclic as for any element (a,b) we have $3(a,b) = (0,0)$– the biggest cyclic subgroup has order 3.

This suggests how we can find the order of an element of a product $\prod_{i=1}^m \mathbb{Z}_{n_i}$ of cyclic groups. Recall that the *least common multiple* $\operatorname{lcm}(r_1, \ldots, r_m)$ *of a set of integers* $\{r_1, r_2, \ldots, r_m\}$ *is the smallest positive integer* n *such that* r_i *divides* n *for all* i.

Fact 11.4. *Let* $a = (a_1, \ldots, a_m) \in G = \prod_{i=1}^n \mathbb{Z}_{n_i}$. *If* a_i *has finite order* r_i *in* $\mathbb{Z}_{n_i}$ *for each* i, *then* a *has order* $\operatorname{lcm}(r_1, \ldots, r_m)$ *in* G.

Proof. Assume that $a = (a_1, \ldots, a_m) \in G$ and that a_i has order r_i in $\mathbb{Z}_{n_i}$. We must show that the least positive integer d such that $da = 0$ is $\operatorname{lcm}(r_1, \ldots, r_m)$. So suppose that $da = 0$. This means that

$$(0, \ldots, 0) = e = da = d(a_1, \ldots, a_m) = (da_1, \ldots, da_m)$$

meaning that for each i, d divides the order r_i of a_i in $\mathbb{Z}_{n_i}$. Thus d divides the least common multiple $\operatorname{lcm}(r_1, \ldots, r_m)$ of the orders of the a_i. On the other hand, if $d = \operatorname{lcm}(r_1, \ldots, r_m)$ then $da_i = 0$ in $\mathbb{Z}_{n_i}$ for all i, and so $da = 0$. □

Problem 11.1. Find the order of $(8, 4, 10)$ in $\mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$.

In particular, we get that the element $(1, \ldots, 1)$ in $\prod_{i=1}^m \mathbb{Z}_{n_i}$ has order $\operatorname{lcm}(n_1, \ldots, n_m)$, so it generates the group if the n_i are pairwise relatively prime.

This allows us to prove the following.

Theorem 11.5. *The group* $G = \prod_{i=1,\ldots d} \mathbb{Z}_{n_i}$ *is cyclic, (and so isomorphic to* $\mathbb{Z}_n$ *where* $n = \prod_{i=1}^d n_i$*) if and only if the numbers* $n_1, \ldots, n_d$ *are pairwise relatively prime.*

Proof. We have already argued that if the n_i are relatively prime then $(1, 1, \ldots, 1)$ generates G, so it is cyclic. On the other hand, if the n_i are not relatively prime then $d := \operatorname{lcm}(n_1, \ldots, n_m) < n$. For any element $(a_1, \ldots, a_m)$ we thus have that $d(a_1, \ldots, a_n) = (da_1, \ldots, da_n) = (0, \ldots, 0)$. And so the order of any element is less than n. So G is not cyclic. □

This tells us that if n has prime decomposition

$$n = p_1^{e_1} p_2^{e_2} \ldots, p_d^{e_d}$$

then the cyclic group of order n can be written as

$$\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{e_1}} \times \mathbb{Z}_{p_2^{e_2}} \times \cdots \times \mathbb{Z}_{p_d^{e_d}},$$

for distinct primes p_i. Insisting that the primes are ordered $p_1 < p_2 < \cdots < p_d$, this gives a sort of 'unique factorisation' finite cyclic groups.

Problem 11.2. Show that $\mathbb{Z}_9 \times \mathbb{Z}_5 \cong \mathbb{Z}_5 \times \mathbb{Z}_9$.

So what is the point of this? It allows us to decide, say, if the group $\mathbb{Z}_6 \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$ having $2^6 3^3 5$ elements is isomorphic to the group $\mathbb{Z}_{12} \times \mathbb{Z}_{18} \times \mathbb{Z}_{40}$ which has the same number of elements. Using the theorem we have that

$$\begin{aligned} \mathbb{Z}_6 \times \mathbb{Z}_{60} \times \mathbb{Z}_{24} &\cong (\mathbb{Z}_3 \times \mathbb{Z}_4) \times (\mathbb{Z}_3 \times \mathbb{Z}_2 \times \mathbb{Z}_5) \times (\mathbb{Z}_3 \times \mathbb{Z}_8) \\ &\cong \mathbb{Z}_2 \times \mathbb{Z}_3^3 \times \mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_8 \end{aligned}$$

Problem 11.3. Factor $\mathbb{Z}_{12} \times \mathbb{Z}_{18} \times \mathbb{Z}_{40}$ and decide if it is isomorphic to $\mathbb{Z}_6 \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$. If they are different find another product of cyclic groups with the same order that is isomorphic to one of them; if they are isomorphic, find another product of cyclic groups of the same order that is not isomorphic to them.

In general we will want to decide if two groups are isomorphic, and now we have a nice way to do it, as long as the groups are (finite) products of finite cyclic groups. Writing a group as a product of cyclic groups has other benefits; we feel we have a pretty good understanding of the group. The product $G = \prod_{i=1}^m \mathbb{Z}_{n_i}$ has a generating set consisting the elements

$$(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, 0, \ldots, 0, 1).$$

This is still true even when some of the factors are $\mathbb{Z}$. Though this set is reminiscent of a basis, and like a basis, it is a generating set, be careful not to give it too much credit. Unlike a basis, it need not be a smallest generating set. We saw that $(1, 1)$ generated $\mathbb{Z}_2 \times \mathbb{Z}_3$. The ith element $s_i = (0, 0, 0, \ldots, 0, 1, 0, \ldots, 0)$ of this set generates subgroup $\overline{\mathbb{Z}}_{n_i} = \{as_i \mid a \in \mathbb{Z}_{n_i}\}$ of $G = \prod_{i=1}^m \mathbb{Z}_{n_i}$ that is clearly isomorphic to $\mathbb{Z}_{n_i}$. When we view G as a product of the subgroups $\overline{\mathbb{Z}}_{n_i}$, we call it an *internal direct product*. When we view it as a product of the Z_{n_i}, we call it an *external direct product*.

We will not prove the following theorem, but it allows us to write any finitely generated abelian group as a product of cyclic groups. So it is good to know.

A group is *finitely generated* if it is generated by some finite subset. Notice that the additive group $\mathbb{Q}$ is not finitely generated. For any finite set of elements, we can write the elements over a common denominator d, and the smallest element we can generate with has absolute value at least $1/d$.

Theorem 11.6 (Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Every finitely generated abelian group is isomorphic to a (finite) direct product of cyclic groups. That is, every such group is isomorphic to a group of the form*

$$\mathbb{Z}_{p_1{}^{r_1}} \times \mathbb{Z}_{p_2{}^{r_2}} \cdots \times \mathbb{Z}_{p_n{}^{r_n}} \times \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$$

where the p_i are primes, not necessarily distinct, and the r_i are positive integers. The product is unique up to the order of the factors.

The number of factors $\mathbb{Z}$ in this representation of a group G is the *Betti number* of G. It is someone's favourite number.

Problem 11.4. Show that there are, up to isomorphism, exactly six different abelian groups of order 360.

We finish off with a couple applications of the Fundamental Theorem.

Theorem 11.7. *If m divides the order of a finite abelian group G, then G has a subgroup of order m.*

Proof. Where $G = \mathbb{Z}_{p_1{}^{r_1}} \times \mathbb{Z}_{p_2{}^{r_2}} \cdots \times \mathbb{Z}_{p_n{}^{r_n}}$ the fact that m divides $|G|$ means that we can write m as $p_1^{s_1} \cdot \ldots p_n^{s_n}$ for some integers S_i satisfying $0 \leq s_i \leq r_i$. As the element $p_i^{r_i - s_i}$ generates a subgroup of $\mathbb{Z}_{p_i}$ of order $p_i^{s_i}$, the subgroup of G generated by the set

$$\{(p_1^{r_1-s_1}, 0, 0 \ldots 0), (0, p_2^{r_2-s_2}, 0, \ldots, 0), \ldots, (0, 0, \ldots, 0, p_n^{r_n-s_n})\}$$

generates a subgroup of G of order $p_1^{s_1} \cdot \ldots p_n^{s_n} = m$. □

Problem 11.5. Show that if m is square-free (not divisible by any square greater than 1) then an group of order m is cyclic.

Problems from the Text

Section 11: 6, 7, 17, 18, 26, **27**, **30**, **32**, 33, 34, 35, **36**, 38, 39, 40, 41, 45, 46, 47, **50, 51**, 52

13 Homomorphisms

We've mentioned them before, but for groups G and G' a *homomorphism* is a map $\phi : G \to G'$ that satisfies the homomorphism property:

$$\forall a, b \in G \qquad \phi(ab) = \phi(a)\phi(b).$$

Recall that if ϕ is a bijective homomorphism then it is an isomorphism. While isomorphisms preserve all of the structural properties of a group. A homomorphism will throw away some of them, it simplifies the picture of the group. But we will see that it usually preserves some the properties too.

We have seen all sorts of homomorphisms in our math careers.

Example 13.1. A real vector space $\mathbb{R}^n$ is a group under addition. Any linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$ of vector spaces has, in particular, that $T(u+v) = T(u) + T(v)$, so is a homomorpism of these additive groups.

Example 13.2. The set of invertible $n \times n$ real matrices is a group $GL_n(\mathbb{R})$ under matrix multiplication. The determinant maps it to the multiplicative group $\mathbb{R}^+$. It is a homomorpism.

Example 13.3. Where $F(\mathbb{R})$ is the additive group of real functions, and c is a constant, the *evaluation homomorphism* $\phi_c : F(\mathbb{R}) \to \mathbb{R}$ to the additive group $\mathbb{R}$ that takes a function f to

$$\phi_c(f) = f(c)$$

is a homomorphism. Indeed

$$\phi_c(f + g) = [f + g](c) = f(c) + g(c) = \phi_c(f) + \phi_c(g).$$

We will use this last example when we get to polynomials over fields. The other two examples above, we don't really use except in examples of other concepts. How about some examples of homomorphisms that we will use a bit more often in our algebra class?

Example 13.4. For an integer n the map $\phi : \mathbb{Z} \to \mathbb{Z} : z \mapsto nz$ is a homomorphism. It certainly maps to $\mathbb{Z}$, and $\phi(a+b) = n(a+b) = na+nb = \phi(a)+\phi(b)$. Is the map $\mathbb{Z}^* \to \mathbb{Z}^* : z \mapsto nz$ of the multiplicative group a homomorphism for all integers n?

Example 13.5. For every integer $n \geq 2$, the map $\mathbb{Z} \to \mathbb{Z}_n : z \mapsto z \mod n$ and if n is prime the map $\mathbb{Z}^* \to \mathbb{Z}_n^* : z \mapsto z \mod n$ of multiplicative groups, is a homomorphism.

Example 13.6. Let $G = \prod_{i=1}^n G_i$ be a direct product of groups. For each i, the *projection map*

$$\pi_i : G \to G_i : (g_1, \ldots, g_n) \mapsto g_i$$

is a homomorphism. Indeed

$$\begin{aligned}\pi_i((g_1,\ldots,g_n)(g_1',\ldots,g_n')) &= \pi_i((g_1g_2',\ldots,g_ng_n') \\ &= g_ig_i' = \pi_i((g_1,\ldots,g_n))\pi_i((g_1',\ldots,g_n')).\end{aligned}$$

The *injection*

$$\iota_i : G_i \to \prod_{i=1}^{n} G_i : g \mapsto (e,e,\ldots,e,g,e,\ldots,e)$$

whose image is $\overline{G}_i \leq G$ is also a homomorphism.

Properties of homomorphisms

If $\phi : A \to B$ and $\psi : B \to C$ are homomorphisms, then so is $\psi \circ \phi : A \to C$. Indeed, we have

$$\begin{aligned}[\psi \circ \phi](a+a') &= \psi(\phi(a+a')) = \psi(\phi(a)+\phi(a')) \\ &= \psi(\phi(a)) + \psi(\phi(a')) = [\psi \circ \phi](a) + [\psi \circ \phi](a').\end{aligned}$$

We prove some of the following for isomorphisms.

Theorem 13.7. *For any homomorphism $\phi : G \to H$, the following hold.*

i). The element $\phi(e)$ is the identity of H.

ii). For any $g \in G$ we have $\phi(g^{-1}) = \phi(g)^{-1}$.

iii). For any subgroup G' of G, $\phi[G']$ is a subgroup of H.

iv). For any subgroup H' of H, $\phi^{-1}[H']$ is a subgroup of G.

Proof. See text, or, actually...

Problem 13.1. Prove this. □

Homomorphisms 'push' a lot of other properties.

Example 13.8. Given a homomomorpism $\phi : G \to H$, if G is abelian, then so is the subgroup $H' := \phi[G]$ of H. In particular, if ϕ is surjective, then H must also be abelian. Indeed let $a, b \in H'$. There are $f, g \in G$ such that $\phi(f) = a$ and $\phi(g) = b$. So

$$ab = \phi(f)\phi(g) = \phi(fg) = \phi(gf) = \phi(g)\phi(f) = ba.$$

This allows us to say that there can be no surjective homomorphism from a cyclic group to, say, S_3. Any homomorphism must map to one of the abelian subgroups. In the other direction non-abelian groups can map onto abelian groups.

Example 13.9. The map $\phi : S_n \to \mathbb{Z}_2$ that takes the set A_n of even permutations to 0, and all odd permutations to 1, is a homomorphism. To show this we must show that $\phi(\sigma\tau) = \phi(\sigma) +_2 \phi(\tau)$. If σ and τ are both even, or both odd, then their composition is even, and this becomes $0 = i +_2 i$ for some $i \in \mathbb{Z}_2$ which is true. If one is even and one is odd then it becomes $1 = i +_2 (1 - i)$ for some $i \in \mathbb{Z}_2$, which is again true.

The kernel

For a homomorphism $\phi : G \to H$ of groups we just saw that $\phi^{-1}[H']$ is a subgroup of G. There is a special case of this that we will use so much that it gets a special name. The preimage $\phi^{-1}[\{e\}]$ of the identity– the subgroup of all things that 'map to 0', is the *kernel,* $\ker\phi$*, of* ϕ.

We have seen in other classes that kernels are important. A linear transformation from linear algebra is a homomorphism of real groups. Its kernel is the null-space. We think of this part of G as 'collapsing to 0' under the homomorphism. At the same time, cosets of the kernel can be viewed as collapsing to a single element. Let's make this rigorous.

Theorem 13.10. *Let $G \to G'$ be a homomorphism, and let $H = \ker\phi$. For any $a \in G$ the set $\phi^{-1}[\phi(a)]$ of elements of G that map to the same place as a is the left coset aH of G, and the right coset Ha of G.*

Proof. Any element of aH can be written as ah for some $h \in H$ and so it maps to $\phi(ah) = \phi(a)\phi(h) = \phi(a)e = \phi(a)$. Thus $aH \subset \phi^{-1}[\phi(a)]$. On the other hand, if $\phi(b) = \phi(a)$ then $\phi(ba^{-1}) = \phi(b)\phi(a)^{-1} = e$ meaning that ba^{-1} is in $\ker\phi = H$, so $b = ah$ for some $h \in h$. Thus $b \in aH$. □

This is saying that if H is the kernel of a homomorphism, then its left and right cosets coincide. We saw that this is not true of every subgroup of G, so those that are the kernel of a homomorphism are special. Indeed, a subgroup H of G is *normal* if its left and right cosets coincide– if $aH = Ha$ for all a. This is not an implicit property of H, it depends on G. The theorem above gives us the following.

Corollary 13.11. *The kernel* $\ker\phi$ *of a homomorphism $\phi : G \to G'$ is a normal subgroup of G.*

We saw in Example 13.9 that there is a homomorphism $\phi : S_n \to \mathbb{Z}_2$ with $\ker\phi = A_n$. Not only is this a nice proof that A_n is a subgroup of S_n, it is a proof that it is a normal subgroup of S_n.

Problem 13.2. The absolute value function $|\cdot|$ from C^* to $\mathbb{R}^+$ is a homomorphism as $|z||z'| = |zz'|$. What is the kernel and what are its cosets?

The following is useful in showing that a homomorphism is injective.

Corollary 13.12. *A group homomorphism $\phi : G \to H$ is injective if and only if* $\ker \phi = \{e\}$.

Proof. By definition ϕ is injective if and only if $|\phi^{-1}(\phi(g))| = 1$ for all g and as a the subgroup $\ker \phi = \phi^{-1}(e)$ and its cosets have the same cardinality, this holds by the Theorem if and only if $|\phi^{-1}(e)| = 1$, which is true if and only if $\ker \phi = \{e\}$.

□

Problems from the Text

Section 13: 1-15, **18,23**, 28, **29**, **32**, **33-41(odd)**, 44, 45, **46**, 48, 50

14 Factor Groups

When we have a subgroup $H \leq G$ of some group G, we get a partition of the elements of G into cosets of H. Let $G/H = \{aH \mid a \in G\}$ be the family of these cosets. Sometimes the group operation on G gives us a *natural* group operation on G/H by 'acting on representatives':

$$aH \cdot bH = abH.$$

This is not always well defined, but we will see that it is if H is a normal subgroup of G.

Recall that H is a normal subgroup of G if the left and right cosets coincide:

$$G/H = \{aH \mid a \in G\} = \{Ha \mid a \in G\} =: G/_R H$$

Because H is a group, we have that $a \in aH$ and $a \in Ha$, so as G/H and $G/_R H$ are partitions of G, the fact that $G/H = G/_R H$ means that $aH = Ha$ for all H. (This is a good thing, too. It is easy to interpret 'the left and right cosets coincide' to mean exactly this.)

That $aH = Ha$ holds does not mean that $ah = ha$ for all $h \in H$, but it does mean that for all $h \in H$ there is some $h^* \in H$ such that $ah = h^*a$. So it almost lets us commute a and h. We will do this a lot.

Theorem 14.1. *For a subgroup H of a group G, the operation on the set G/H of left cosets is defined by*

$$aH \cdot bH = abH$$

is well defined if and only if H is normal. If it is well defined, then G/H is a group under this operation.

Proof. Assume that H is normal, we show that the operation is well defined. This means that it cannot depend on the representation of the coset aH. That is, take any a' in H, so $a'H = aH$. We have to show that $a'bH = abH$ so that $a'H \cdot bH = aH \cdot bH$ is independent of the representative a'. Well, as $a' \in H$ we have that $a' = ah$ for some $h \in H$, and as H is normal $hb = bh*$ for some $h* \in H$. So

$$a'bH = ahbH = abh * H = abH.$$

Similarily we can show for any $b' \in H$ that $ab'H = abH$, and so the operation is well defined.

Now, assume that the operation is well defined. We show that $aH = Ha$ for any $a \in G$, showing that H is normal. Let $x \in aH$, it is enough to show that $x \in Ha$. As $x \in aH$ we have $xH = aH$ and so $xa^{-1}H = xHa^{-1}H = aHa^{-1}H = eH = H$ meaning that $xa^{-1} = h \in H$. So $x = ha$ is in Ha. Thus $aH \subset Ha$ and a similar proof shows $aH \supset Ha$, and so $aH = Ha$.

To see now that G/H is a group, observe that the operation is clearly closed. That eH is the identity element, and that $a^{-1}H$ is the inverse of aH is now easy, and so G/H is a group. $\square$

The group G/H is called a *factor group*, or *quotient group*, *of G modulo H*.

Example 14.2. The group $\mathbb{Z}_{15}$ is abelian, so its subgroup $\langle 3\rangle = \{0, 3, 6, 9, 12\}$ is normal. The set $\mathbb{Z}_{15}/\langle 3\rangle$ of cosets

$$0+\langle 3\rangle = \{0, 3, 6, 9, 12\}, \qquad 1+\langle 3\rangle = \{1, 4, 7, 10, 13\} \qquad 2+\langle 3\rangle = \{2, 5, 8, 11, 14\}$$

is a group with addition defined by acting on representatives. Coset represenation is a bit unwieldy, so to save some writing it is not uncommon to use class notation for cosets, and write $1 + \langle 3\rangle$ as $[1] = [4] = [7]$, etc. We can use any representative to denote the class, and the group action works on represetatives:

$$[3] + [7] = [3 + 7] = [10] = [1].$$

Example 14.3. The group $\langle(1, 0)\rangle = \overline{\mathbb{Z}}_2$ is normal in the abelian group $\mathbb{Z}_2 \times \mathbb{Z}_3 = \overline{\mathbb{Z}}_2 \times \overline{\mathbb{Z}}_3$ and so we have the factor group $\overline{\mathbb{Z}}_2 \times \overline{\mathbb{Z}}_3/\langle(1, 0)\rangle$. It has elements

$$(0, 0) + \langle(1, 0)\rangle \qquad (0, 1) + \langle(1, 0)\rangle \qquad (0, 2) + \langle(1, 0)\rangle$$

and is clearly isomorphic to the subgroup $\overline{\mathbb{Z}}_3$. So we have that

$$(\overline{\mathbb{Z}}_2 \times \overline{\mathbb{Z}}_3)/\overline{\mathbb{Z}}_2 \cong \overline{\mathbb{Z}}_3.$$

This perhaps explains the name 'factor group'. We have collapsed one factor of the product group, and left the other factor.

Consider this example for a group that is a direct product, it is perhaps tempting to guess, for any normal subgroup N of G, that $G \cong (G/N \times N)$. But this is not true, as the following example shows. (And indeed if it were true, then normal subgroups would not be as interesting, as they would then just be factors of direct products.) This is why many people prefer the name of 'quotient group' to 'factor group'.

Example 14.4. We saw that the map $\phi : \mathbb{Z} \to \mathbb{Z}_n : z \mapsto z \mod n$ is a homomorphism, and its kernel is clearly $n\mathbb{Z}$. So $n\mathbb{Z}$ is normal in $\mathbb{Z}$, (of course it is, $\mathbb{Z}$ is abelian), and

$$\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}.$$

Note that $\mathbb{Z}_n$ can never be viewed as a factor of some product isomorphic to $\mathbb{Z}$, as it has elements of finite order, while $\mathbb{Z}$ does not.

This example is a prototypical example of a quotient group and often the terminology we know from the integers modulo n is carried over to other quotient groups: G/N is the quotient of G *modulo* N; further, if $gN = g'N$ we say that g and g' are *congruent modulo* N.'

Now, for any homomorphism $\phi : G \to G'$ we saw that the kernel of ϕ is normal in G, so the quotient group $G/\ker\phi$ is well defined. Well, this quotient group is isomorphic to the image of G in G'.

Theorem 14.5. *Let $\phi : G \to G'$ be a homomorphism, and let $N = \ker\phi$. The map $\mu : G/N \to \phi[N] \leq G'$ defined by*

$$\mu(gN) = \phi(g)$$

is an isomorphism.

Proof. First we show that μ is well-defined. If $gN = g'N$ then $g' = gh$ for some $h \in N$, and so

$$\mu(g'N) = \phi(g') = \phi(gh) = \phi(g)\phi(h) = \phi(g)e = \phi(g) = \mu(gN),$$

as needed. That μ is onto $\phi[N]$ is clear, and that to see that it is injective, it is enough to observe that its kernel contains only the element N. That ϕ satisfies the homomorphism property is also clear. □

There is one more part to the picture. From a homomorphism $\phi : G \to G'$ defines a normal subgroup $N = \ker\phi$ of G and so a quotient group G/N of G. But we get a quotient group G/N for any normal subgroup of G. This allows us to show that any normal subgroup N of G is the kernel of some homomorphism of G, indeed, the homomorphism to G/N.

Theorem 14.6. *Let N be a normal subgroup of G. The map $\gamma : G \to G/N$ defined by $\gamma(x) = xN$ is a homomorphism with kernel N.*

Proof. For g and g' in G we have

$$\gamma(gg') = gg'N = gNg'N = \gamma(g)\gamma(g')$$

so γ is a homomorphism, and that the kernel is N is clear, as N is the identity element of G/N. □

Given a homomorphism $\phi : G \to G'$ with $N = \ker\phi$ we saw by Theorem 14.5 that the map $\mu : G/N \to \phi[G]$ defined by $\mu(gN) = \phi(g)$ is an isomorphism, we call it the canonical isomorphism. Then by Theorem 14.6 we say that $\gamma : G \to G/N$ defined by $\gamma(g) = gN$ is a homomorphish; we call it the *canonical quotient*. Putting these together, we get that $\mu(\gamma(g)) = \mu(gN) = \phi(g)$. That is to say, ϕ *factors as* $\phi = \mu \circ \gamma$. This gives us the following.

Theorem 14.7 (The (Fundamental) Homomorphism Theorem)**.** *For a group homomorphism $\phi : G \to G'$ with kernel $N = \ker\phi$ the map $\mu : G/N \to \phi[G]$ defined by $\mu(gN) = \phi(g)$ is an isomorphism, the map $\gamma : G \to G/N$ defined by $\gamma(g) = gN$ is a homomorphism, and $\phi = \gamma \circ \mu$.*

This gives us, for any group G, the useful correspondences

$$\text{normal subgroups} \iff \text{factor groups} \iff \text{homomorphic images.}$$

Example 14.8. The group $(Z_4 \times Z_2)$ is abelian so any subgroup is normal, in particular the subgroup $H = \{0\} \times Z_2$ is, so we have a factor group $(Z_4 \times Z_2)/H$. What is this group? Well, as H is the kernel of the projection map $\pi : Z_4 \times Z_2 \to Z_4$ defined by $\pi(a, b) = a$ the factor group $(Z_4 \times Z_2)/H$ is isomorphic to $\pi[Z_4 \times Z_2] = Z_4$.

Normal subgroups and inner automorphisms

By definition H is normal in G if the left cosets and the right cosets coincides, and we saw that this is true if and only if $aH = Ha$ for all $a \in G$. There are a couple of equivalent characterisations of normal subgraphs that will be useful in the exercises and in the sequel.

Problem 14.1. Show that for a subgroup H of G the following are equivalent.

i). $gH = Hg$ for all $g \in G$

ii). $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$

iii). $gHg^{-1} = H$ for all $g \in G$.

Problem 14.2. Show that for any group G and any element g of G the map

$$i_g : G \to G : a \mapsto gag^{-1}$$

is an isomorphism.

An isomorphism of a group to itself is an *automorphism*. This automorphism i_g is called an *inner automorphism of* G. It is said to *conjugate an element* a *by* g.

The characterisation of a subgroup H as normal if $gHg^{-1} = H$ for all $g \in G$ means that H is *invariant* under i_g for all g. (Again, this does not necessarily mean that $ghg^{-1} = h$ for all $g \in G$ and $h \in H$.) So the normal subgroups of G are exactly the groups that are invariant under all inner automorphisms. If H is not normal then there are other subgroups K of G isomorphic to H, such that $K = i_g(H)$ for some inner automorphism of G. Such groups are *conjugate subgroups* of H in G.

Problems from the Text

Section 14: 6, **13**, **17-19**, 21, **23**, **26**, 27-29, **30**, 31-33, **34**, 35, 36, 39

15 Factor Group Computations and Simple Groups

We look at several examples of normal subgroups of groups and *compute* their factor groups. When possible, 'compute' means we write them according to the representation assured us by the Fundamental Theorem of Finitely Generated Abelian Group. For other groups, the meaning of *compute* will be made more clear in Chapters 32-34.

We start with the easiest normal subgoups– the trivial one and the non-proper one.

Example 15.1. The *trivial* subgroup $N = \langle e \rangle$ of any group G is always normal. The factor group G/N has elements gN for all $g \in G$, each coset has order 1. The group N is the kernal of the map $\mathrm{id} : G \to G : g \mapsto g$, and so the canonoical isomorphism is $\mu : G/N \to \mathrm{id}[G] = G$.

Example 15.2. The group G is always a normal subgroup of G, so G/G is a factor group. As G is the kernel of the homomorphism of G to the trivial group $\langle e \rangle$, G/G is isomorphic to $\langle e \rangle$.

The trivial and non-proper normal subgroups of a group G are not really interesting, and now that we have acknowledged them, we try to keep to non-trivial proper normal subgroups N of a group G.

Notice that any subgroup N of G such that $(G : N) = 2$ must be normal, as N itself is always both a right and left coset, and as $(G : N) = 2$ the only other coset, left or right, is $G \setminus N$. So trivially the left and right cosets coincide.

Example 15.3. The subgroup A_n of S_n has index $(S_n : A_n) = 2$, so is normal. Its factor group S_n/A_n has two elements A_n and $(1, 2)A_n$ so is isomorphic to $\mathbb{Z}_2$.

The following example was promised a while ago. It shows that the converse of Legrange's Theorem need not hold for non-abelian groups. That is we show that A_4, which has order 12 does not have a subgroup of order 6, even though 6 divides 12.

Example 15.4. Assume that A_4 has a subgroup N of order 6. As this has index 2 it is normal, and so there is a factor group A_4/N of order 2. This factor group has two elements, (the cosets N and σN for some transposition σ) and as the subgroup is isomorphic $\mathbb{Z}_2$, we have that $NN = N$ and $(\sigma N)(\sigma N) = N$. Any $\tau \in A_4$ is in one of these cosets, so $(\tau^2 N) = (\tau N)(\tau N) = N$ meaning that the square τ^2 of any element τ of A_4, is in N. Now any three cycle (a, b, c) of A_4 can be written as $(a, c, b)^2$ so is in N, but there are eight such elements in A_4. This is impossible, as $|N| = 6$.

In the last section we computed $(\mathbb{Z}_2 \times \mathbb{Z}_3)/\langle (1, 0) \rangle \cong \mathbb{Z}_3$.

Example 15.5. Compute the group $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(2,3)\rangle$. Did you say it is

$$\mathbb{Z}_4/\langle 2\rangle \times \mathbb{Z}_6/\langle 3\rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_3?$$

It doesn't work this way. We don't break the element $(2,3)$ up like that. We have that

$$\langle(2,3)\rangle = \{(0,0),(2,3)\}.$$

The cosets of $\langle(2,3)\rangle$ all have size 2, and $\mathbb{Z}_4 \times \mathbb{Z}_6$ has order 24, so the factor group has order 12. Having order 12, the factor group must be $\mathbb{Z}_4 \times Z_3$ or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times Z_3$. How can we tell which one it is? The first of these has an element of order 4, but the second doesn't. Can you find an element of order 4 in $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(2,3)\rangle$? Sure the element $[(1,0)]$ generates $\{[(1,0)],[(2,0)],[(3,0)],[(0,0)]\}$. Keep adding it to itself until we get an element of $\langle(2,3)\rangle$. In fact, you might have found that $[(1,1)]$ has order 12. The group is $\mathbb{Z}_{12} = \mathbb{Z}_4 \times \mathbb{Z}_3$.

The following generalises Example 14.3 where we saw $(\mathbb{Z}_2 \times \mathbb{Z}_3)/\overline{\mathbb{Z}}_2 \cong \mathbb{Z}_3$.

Theorem 15.6. *Let $H \times K$ be a direct product of groups. Then $\overline{H} = \{(h,e) \mid h \in H\}$ is a normal subgroup and $(H \times K)/\overline{H} \cong K$.*

Proof. The projection $\pi_2 : H \times K \to K : (h,k) \mapsto k$ is a homomorphism with kernel $\overline{H}$ and so $\overline{H}$ is normal. By the homomorphism theorem, the map $\mu : (H \times K)/\overline{H} \to \pi_2[H \times K] = K$ is an isomorphism. □

The following applies to Example 14.4 where we saw that $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Theorem 15.7. *A factor group of a cyclic group is cyclic.*

Proof. Where a is a generator of the cyclic group G, aN is a generator of G/N for any normal subgroup N. Indeed as we compute the product $d(aN)$ in G as $(da)N$ we clearly generate all cosets of N. □

Example 15.8. We compute $(\mathbb{Z} \times \mathbb{Z})/\langle(1,1)\rangle$. The subgroup $\langle(1,1)\rangle$ consists of the points $\{(z,z) \mid z \in \mathbb{Z}\}$ on the positive diagonal through $(0,0)$ of the integer plane. Its coset $(n,0) + \langle(1,1)\rangle$ is the positive diagonal through $(n,0)$. Factoring out by these diagonals collapses $(0,n) + \langle(1,1)\rangle$ to the point $(0,1)$ so $(\mathbb{Z} \times \mathbb{Z})/\langle(1,1)\rangle$ has representitives

$$\ldots,[(-2,0)],[(-1,0)],[(0,0)],[(1,0)],[(2,0)],\ldots.$$

This is isomorphic to $\mathbb{Z}$.

Problem 15.1. Show that $(\mathbb{Z} \times \mathbb{Z})/\langle(1,1)\rangle \cong \mathbb{Z}$ by finding a homomorphism from $\mathbb{Z} \times \mathbb{Z}$ to $\mathbb{Z}$ with kernel $\langle(1,1)\rangle$, and then using the homorphism theorem.

Simple Groups

A group is *simple* if it has no non-trivial proper normal subgroup. We know that the only abelian simple groups are cyclic, and this fact is basic in the proof of the fundamental theorem of finitely generated abelian groups. The classification of non-abelian simple groups is not as easy.

You will show in Exercise 15.39 that the alternating group A_n is simple for $n \geq 5$. The non-abelian simple groups have been classified, the classification took 30 years, being completed in 1980, and consists of over 5000 journal pages. The text says more about this.

We won't list all of these simple groups, but we will look at how non-abelian groups can be expressed in terms of them. That is, we will look at finding the normal subgroups of non-simple groups.

Strengthening our earlier result that said homomorphisms preserve subgroups the following, which you show in Exercises 15.35 and 15.36, says that homomorphisms preserve normal subgroups.

Theorem 15.9. *Let $\phi : G \to G'$ be a homomorphism. If N is a normal subgroup of G then $\phi[N]$ is normal in $\phi[G]$. If N' is a normal subgroup of G' then $\phi^{-1}[N']$ is normal in G. (Note here that $\phi[N]$ need not be normal in G', we can only say it is normal in $\phi[G]$.)*

Proof. Assume that N is normal in G. To see that $\phi[N]$ is normal in $\phi[G]$ let $\phi(n) \in \phi[N]$ and $\phi(g) \in \phi[G]$. As N is normal we have $gng^{-1} \in N$ and so

$$\phi(g)\phi(n)\phi^{-1}(g) = \phi(gng^{-1}) \in \phi[N],$$

as needed.

We can use almost the same proof in the other direction, and you should try it, but the homomorphism theorem gives us a cute proof here.

If N' is normal in G' then there is a homomorphism $\phi' : G' \to G''$ with kernel N'. Composing, we get a homomorphism $\phi' \circ \phi : G \to G''$ with kernel $\phi^{-1}[N']$, so this is normal. □

The *maximal normal subgroup of a group G* is a proper normal subgroup of G that it not properly contained in an other proper normal subgroup of G.

Theorem 15.10. *Let N be a normal subgroup of G. The factor group G/N is simple if and only if N is maximal.*

Proof. Let $\gamma : G \to G/N$ be the canonical quotient. For both directions we argue the contrapositive.

If G/N is not simple, it has a non-trivial proper normal subgroup H. The preimage $\gamma^{-1}(H)$ is normal in G; proper as $H \leq G/N$ is proper; contains N as $\gamma^{-1}(e) = N$ and $e \in H$; and properly contains N as H is non-trivial. We have showed that G/N being not simple implies that N is not maximal, so if N is maximal then G/N is simple.

For the other direction, assume that N is not maximal. This means there is a proper normal subgroup M of G which properly contains N. We claim that the group $\gamma[M]$ is proper non-trivial normal subgroup of G/N, showing that G/N is not simple. Well $\gamma[M]$ is normal, and as M properly contains N, $\gamma[M]$ is non-trivial. What is left to show is that $\gamma[M]$ is a proper subgroup of G/N. If it isn't then where g is in $G \setminus M$, we have that $\gamma(g) = \gamma(m)$ for some $m \in M$. But then

$$gN = \gamma(g) = \gamma(m) = mN$$

and so $m^{-1}g \in N \subset M$. As M is a group we have then that $g = mm^{-1}g \in M$, contradicting the fact that $g \in G \setminus M$. $\square$

The Center and Commutator Subgroups

To describe non-abelian groups in terms of their simple factor groups, we have to find normal subgroups. There are two main normal subgroups that we can always find in a non-simple group G. The center is the largest abelian subgroup of G. The commutator is the subgroup we have to quotient out to make G abelian.

The *center* of a group G is the subgroup

$$Z(G) := \{a \in G \mid \forall g \in G, ag = ga\}.$$

This is a group. Indeed, let a and b be in $Z(G)$, so $ag = ga$ and $bg = gb$. From this we get

$$a^{-1}bg = a^{-1}gb = a^{-1}gaa^{-1}b = a^{-1}aga^{-1}b = ga^{-1}b,$$

showing that $a^{-1}b$ is also in $Z(G)$. Moreover $Z(G)$ is normal because for any $g \in G$ and $a \in Z(G)$ we have that gag^{-1} is in $Z(G)$; indeed $gag^{-1} = agg^{-1} = a \in Z(G)$.

Problem 15.2. Show that G is abelian if and only if $G = Z(G)$.

The center of G always contains the identity, and if this is all that it contains, we say that the center is *trivial*.

Problem 15.3. Show that S_3 has a trivial center.

The *commutator* subgroup of G is the subgroup

$$\{aba^{-1}b^{-1} \mid a, b \in G\}.$$

Problem 15.4. Show that this is indeed a subgroup.

Fact 15.11. *The commutator C of a group G is normal.*

Proof. For any g in G and any $aba^{-1}b^{-1}$ in C we see that $g(aba^{-1}b^{-1})g^{-1}$ is in C as

$$gaba^{-1}b^{-1}g^{-1} = gaba^{-1}(g^{-1}b^{-1}bg)b^{-1}g^{-1} = [(ga)(b)(a^{-1}g^{-1})(b^{-1})][bgb^{-1}g^{-1}]$$

which is in C. □

So the quotient G/C is well defined. Moreover, for elements a and b of G we have that $aCbC = abC$, and because $b^{-1}a^{-1}ba$ is in C we get that $abC = baC = bCaC$, so $aCbC = bCaC$ showing that G/C is abelian. In fact this is the smallest quotient of G that is abelian.

Theorem 15.12. *Let C be the commutator subgroup of G. For a normal subgroup N of G we have that G/N is abelian if and only if $C \leq N$.*

Proof. If G/N is abelian then where $\gamma : G \to G/N$ is the canonical quotient, for any ab in G we have that

$$\gamma(aba^{-1}b^{-1}) = \gamma(a)\gamma(b)\gamma(a^{-1})\gamma(b) = \gamma(a)\gamma(a^{-1})\gamma(b)\gamma(b^{-1}) = e$$

and so $aba^{-1}b^{-1} \in \ker\gamma = N$, so $C \leq N$.

On the other hand if $C \leq N$ then for any aN, bN in G/N we have that $a^{-1}b^{-1}ab \in N$, and so $bNaN = baN = ba(a^{-1}b^{-1}ab)N = abN = aNbN$, showing that G/N is abelian.

□

Problems from the Text

Section 15: 3,5,10, 13, **14**, **19**, **20**, 21-23, 24-27, 29, **34**, **35-36**, 39, **40**, 42

16 Group action on a set

A G-*action* on a set X is a map $\phi : G \times X \to X : (g, x) \mapsto gx$ such that $ex = x$ for all $x \in X$ and $(gg')x = g(g'x)$ for all $g, g' \in G$ and $x \in x$. We usually omit 'ϕ' and simply say that X is a G-set to mean that there is a G-action on X and its is written as concatencation: gx.

The most obvious example of group action is the following.

Example 16.1. For any set X and any subgroup H of the set S_X of permutations of X, H acts on X by

$$\sigma x = \sigma(x)$$

for all $\sigma \in H$ and $x \in X$.

In fact, this is really the only example.

Theorem 16.2. *Let X be a G-set for a group G. The action $x \mapsto gx$ for any element g is a permutation σ_g of X and the map $\phi : G \to S_X : g \mapsto \sigma_g$ is a homomorphism.*

Proof. As the action $x \mapsto gx$ has an inverse action $x \mapsto g^{-1}x$, it is a bijection, so is a permutation σ_g of X. That ϕ is a homomorphism is immediate as

$$\sigma_{gg'}(x) = (gg')x = g(g'x) = \sigma_g(g'x) = \sigma_g(\sigma_{g'}(x)) = (\sigma_g \circ \sigma_{g'})(x).$$

□

The action of G on a set X is *faithful* if the only element of G that acts as the identity on X is e. The kernel $N = \ker \phi$ of ϕ from the above theorem is the set of elements of G whose action is the identity on X. So this subset is a normal subgroup of G, the factor group G/N also acts on X by

$$(gN)x = gx.$$

The action of gN on X is faithful. So any group action gives a faithful group action by a factor group.

Here are some more examples of group actions.

Example 16.3. G itself is a G-set with G-action simply defined $gx = gx$. The group $G \times G \times G$ is a G-set with G action defined by $g(a, b, c) = (ga, gb, gc)$.

Problem 16.1. Does the same action as in the example above define a G-action on a subgroup H of G? How about an H-action of G?

Example 16.4. Any subgroup H of G acts on G by conjugation. The action of an element $h \in H$ is

$$\sigma_h : G \to G : g \mapsto \sigma_h(g) = hgh^{-1}.$$

(We have to be careful denoting the action this example, using the usual concatenation for the action and the operation in G would have us writing $hg = hgh^{-1}$ which would be confusing.)

The Orbit-Stabilizer Theorem

Given a G-set X and $x \in X$, the sets

$$Gx = \{gx \mid g \in G\} \subset X \qquad \text{and} \qquad G_x = \{g \in G \mid gx = x\} \subset G$$

are the *orbit* and *stabilizer* of x respectively.

Theorem 16.5. *For any G-set X and any $x \in X$, G_x is a subgroup of G.*

Proof. Clearly $e \in G_x$ as $ex = x$. Let g and g' be in G. Then $gg'x = gx = x$ so gg' is in G_x, and to see that g^{-1} is in G_x observe that $x = ex = (g^{-1}g)x = g^{-1}x$. □

The stabilizer G_x is also called the *isotropy subgroup of* x in G. The following is known as the Orbit-Stabilizer Theorem.

Theorem 16.6. *Let X be a G-set and let $x \in X$. The map $g \mapsto gG_x$ is a bijection of Gx to the left cosets of G_x.*

Proof. First we observe that the set $G_{x,y} := \{g \in G \mid gx = y\}$ is a left coset of G_x for any $y \in Gx$. Indeed, take some $g \in G_{x,y}$. For any $h \in G_x$, we have $ghx = gx = y$, and so gh is also in $G_{x,y}$. Thus $G_{x,y} \supseteq gG_x$. On the other hand $g = ge \in gG_x$ so $G_{x,y}$ is the left coset gG_x.

The sets $G_{x,y}$ as y runs over Gx are clearly a partition of G, and so gG_x as gx runs over Gx is also a partition. So the map $g \mapsto gG_x$ is a bijection. □

With Legrange's Theorem, we then get the following.

Corollary 16.7. *Let X be a G-set and let $x \in X$. Then $|Gx| = (G : G_x)$. In particular, if $|G|$ is finite, then $|Gx|$ divides $|G|$.*

For the problems the following definition.

A G-action on a set X is *transitive* if for all $x, y \in X$ there is some $g \in G$ such that $gx = y$.

Problems from the Text

Section 16: 1-3, **6**, **8**, 9, **11**, **12**